Supplementary Appendix 4

Contents

1 Plots of survival curves **1**

2 Probabilistic distributions of 10- and 20-year OS **5**

3 Importance sampling: model diagnostics, parameter distributions, and survival estimates **7**

3.1 Comparisons of parameter estimates with and without external information 7

3.2 Effective sample size plots 9

3.3 Distribution comparison plots 12

3.4 Parameter variability 16

4 Comparison with expertsurv output **16**

4.1 Plots of survival curves over time 16

4.2 Comparisons of parameter distributions 19

4.3 Comparisons of Mean OS 22

4.4 Comparisons of 5-year OS probabilities 23

5 Scenario analysis: alternative specifications of external information **24**

5.1 Description of included scenarios 24

5.2 Results of Scenario analyses: 25

6 Assessment of multivariate normal approximation of joint parameter distributions **31**

1 Plots of survival curves

The below plots show the estimated survival curves (pointwise probabilistic median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, for the survival distributions under consideration. Vertical bars indicate the 95% probability interval for 5 year OS based on external information.

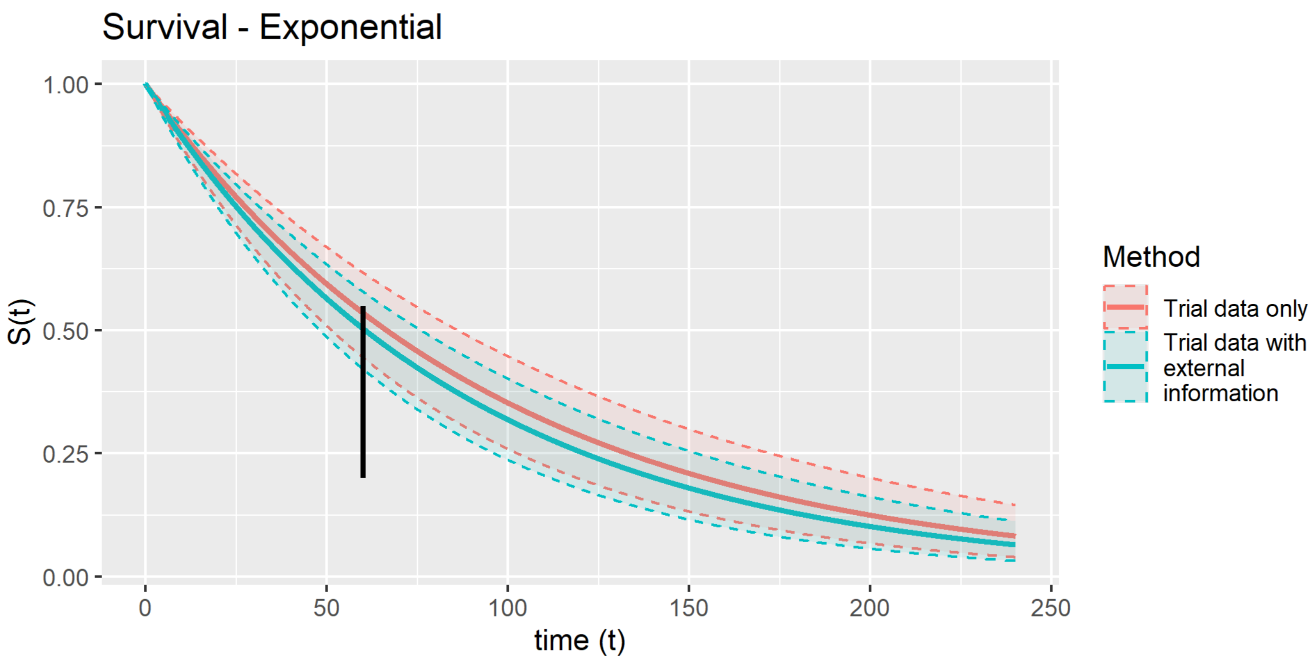

Figure 1: Survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, Exponential distribution. Vertical bar indicates 95% probability interval for 5 year OS based on external information.

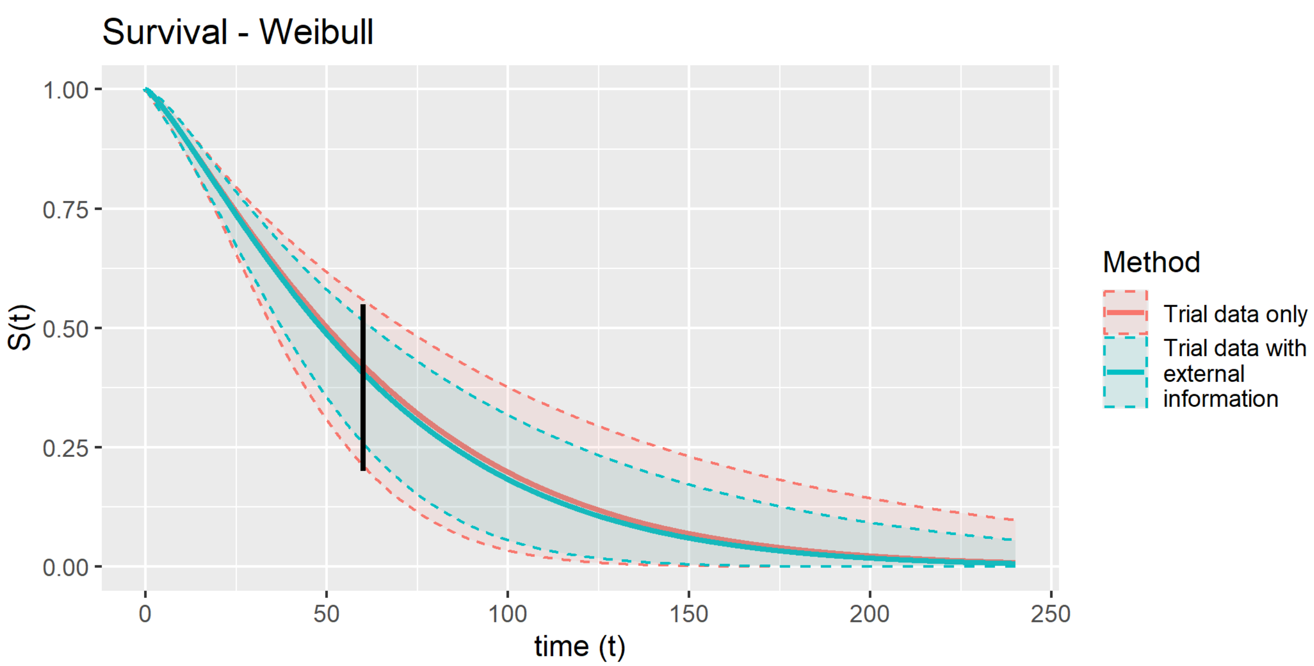

Figure 2: Survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, Weibull distribution. Vertical bar indicates 95% probability interval for 5 year OS based on external information.

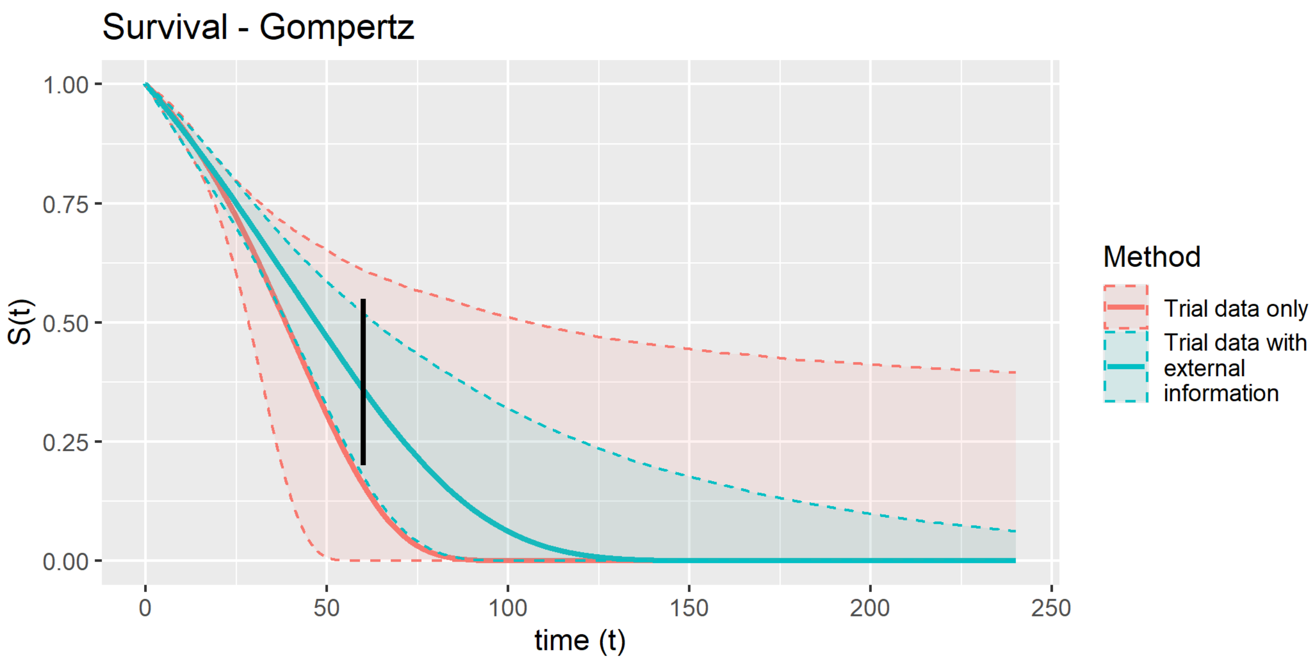

Figure 3: Survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, Gompertz distribution. Vertical bar indicates 95% probability interval for 5 year OS based on external information.

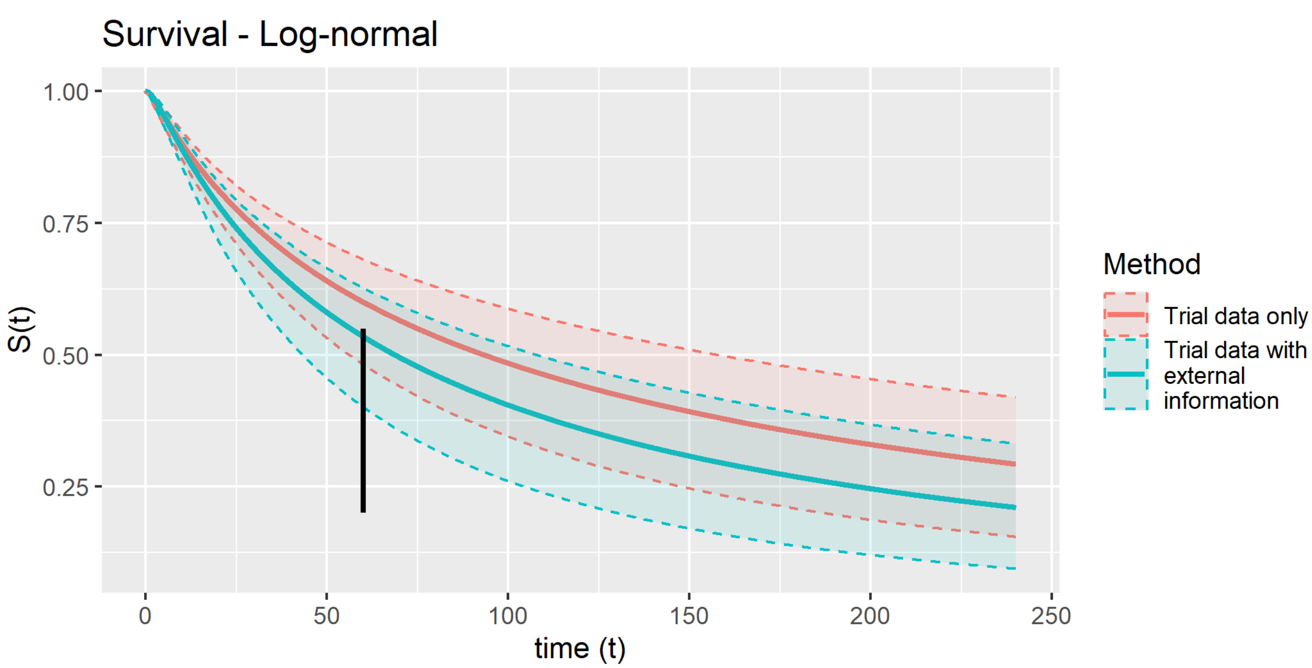

Figure 4: Survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, Log-normal distribution. Vertical bar indicates 95% probability interval for 5 year OS based on external information.

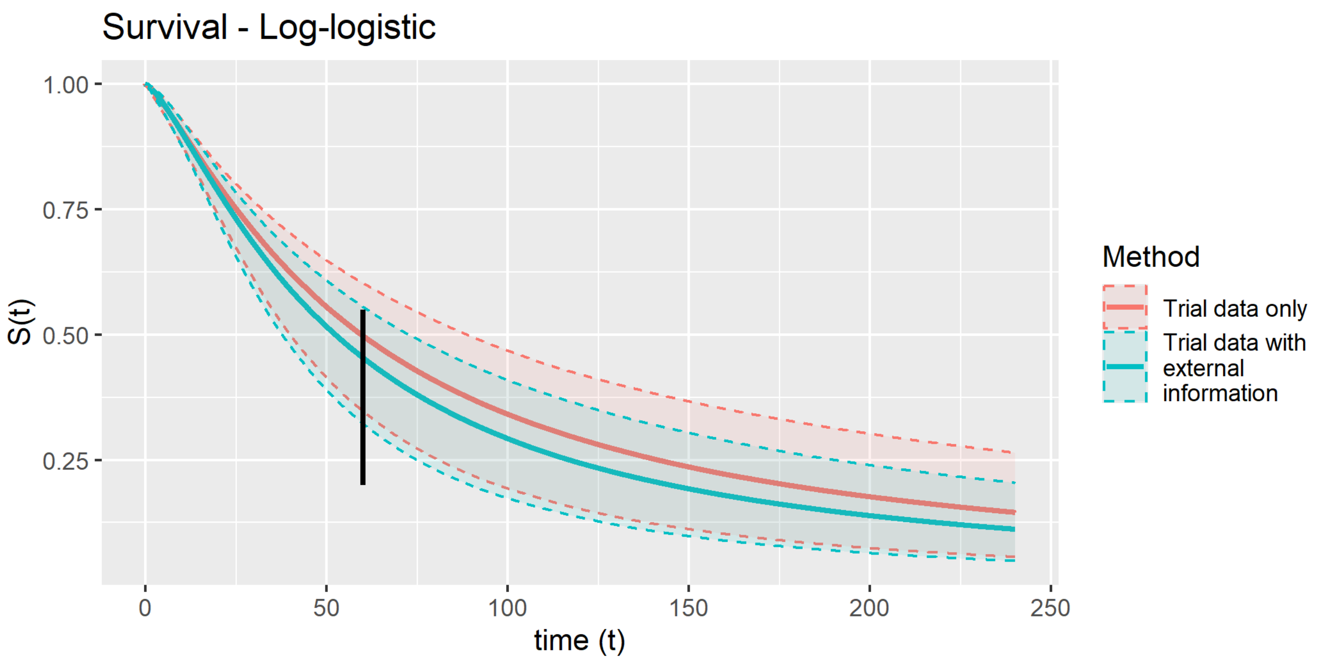

Figure 5: Survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, Log-logistic distribution. Vertical bar indicates 95% probability interval for 5 year OS based on external information.

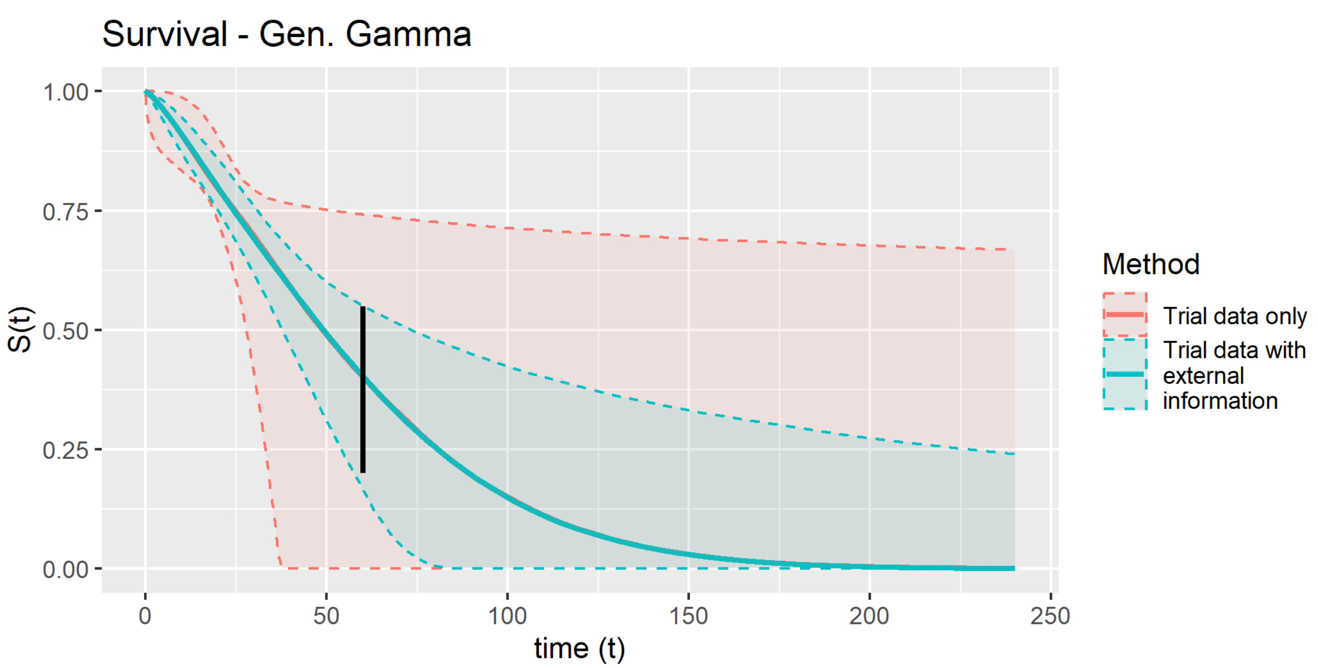

Figure 6: Survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, Gen. Gamma distribution. Vertical bar indicates 95% probability interval for 5 year OS based on external information.

2 Probabilistic distributions of 10- and 20-year OS

Table 1: Summary of probabilistic distributions of 10- and 20-year OS, estimated using Trial Data Only via MLE, and combining Trial Data with external information via the sansIPD (importance sampling) method

		Trial Only			Trial + External			Difference	
Distribution	Timepoint	Mean	Median	95% CrI	Mean	Median	95% CrI	Mean Diff.	Var. Ratio
Exponential	10 years	0.29	0.29	(0.20 to 0.38)	0.25	0.25	(0.18 to 0.33)	-0.03	0.79
Exponential	20 years	0.08	0.08	(0.04 to 0.14)	0.07	0.06	(0.03 to 0.11)	-0.02	0.61
Gen. Gamma	10 years	0.19	0.07	(0.00 to 0.70)	0.11	0.08	(0.00 to 0.38)	-0.08	0.25
Gen. Gamma	20 years	0.12	0.00	(0.00 to 0.66)	0.03	0.00	(0.00 to 0.24)	-0.09	0.11
Gompertz	10 years	0.05	0.00	(0.00 to 0.47)	0.04	0.01	(0.00 to 0.24)	0.00	0.31
Gompertz	20 years	0.02	0.00	(0.00 to 0.39)	0.00	0.00	(0.00 to 0.05)	-0.02	0.07
Log-logistic	10 years	0.29	0.29	(0.15 to 0.43)	0.24	0.24	(0.13 to 0.36)	-0.05	0.70
Log-logistic	20 years	0.15	0.15	(0.05 to 0.27)	0.12	0.11	(0.04 to 0.21)	-0.03	0.58
Log-normal	10 years	0.44	0.44	(0.30 to 0.55)	0.36	0.36	(0.22 to 0.48)	-0.08	1.05
Log-normal	20 years	0.29	0.29	(0.16 to 0.42)	0.21	0.21	(0.10 to 0.34)	-0.08	0.82
Weibull	10 years	0.14	0.13	(0.01 to 0.32)	0.12	0.12	(0.02 to 0.25)	-0.02	0.51
Weibull	20 years	0.02	0.01	(0.00 to 0.10)	0.01	0.01	(0.00 to 0.06)	-0.01	0.30

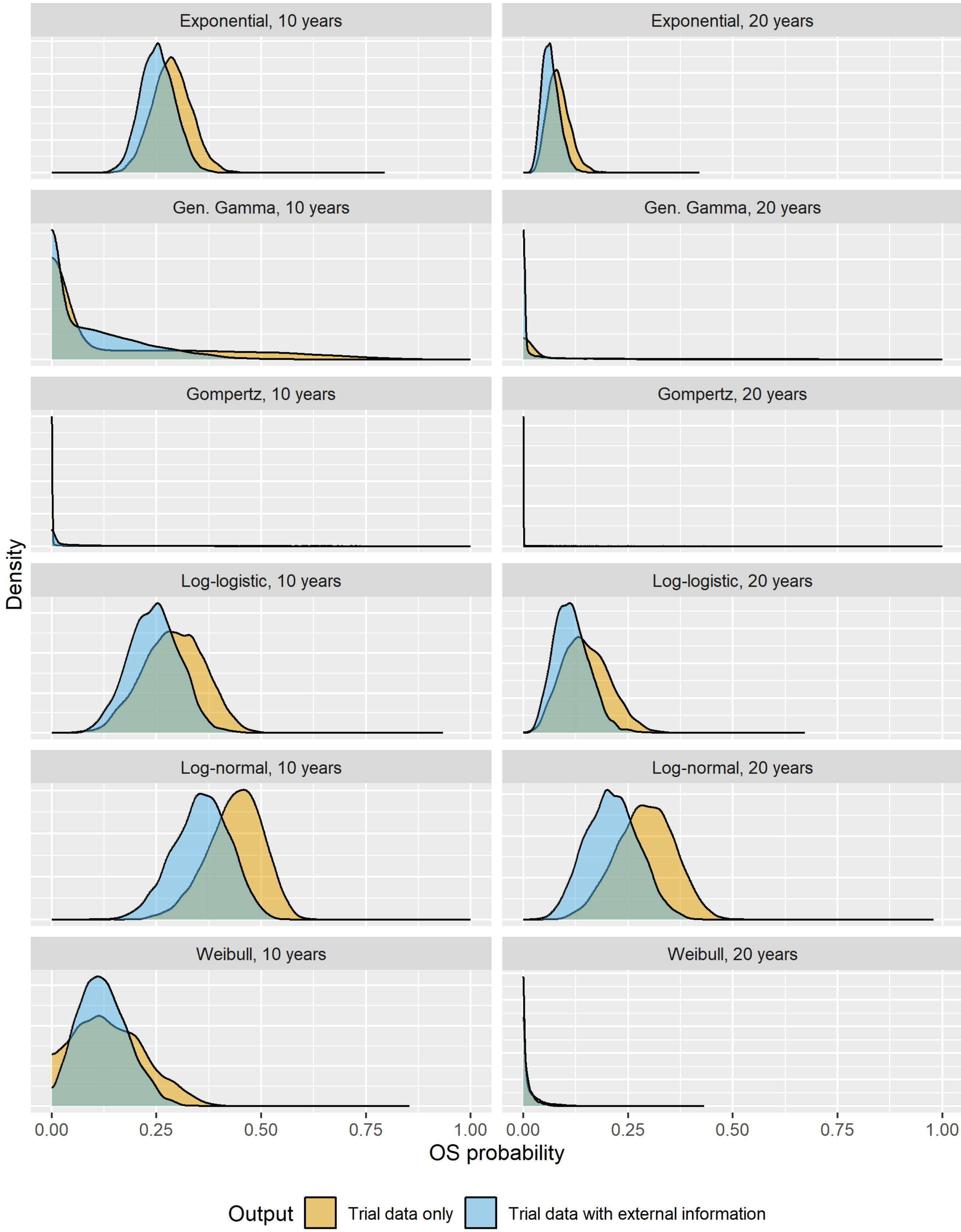

Figure 7: Density plots of probabilistic distributions of 10- and 20-year OS, estimated using Trial Data Only via MLE, and combining Trial Data with external information via the sansIPD (importance sampling) method

3 Importance sampling: model diagnostics, parameter distributions, and survival estimates

The tables and figures in this section show survival curve parameter estimates, covariance structures, importance sampling diagnostics, and (probabilistic) survival curve predictions, obtained from trial data only estimated using maximum likelihood estimation (MLE), and combining trial data with external information using the importance sampling algorithm described in the methods section of the main article and in Appendix 1.

3.1 Comparisons of parameter estimates with and without external information

The below tables compare parameter estimates and covariance matrices obtained from trial data only with those obtained combining trial data with external information, for each of the survival models under consideration.

Table 2: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Exponential distribution.

	x
rate	-4.562874

	x
rate	-4.467088

Table 3: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Exponential distribution.

	rate
rate	0.0169492

	rate
rate	0.0138444

Table 4: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Weibull distribution.

	x
shape	0.2028128
scale	4.2127617

	x
shape	0.217308
scale	4.175548

Table 5: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Weibull distribution.

	shape	scale
shape	0.0147844	-0.0228526
scale	-0.0228526	0.0466214

	shape	scale
shape	0.0091972	-0.0115360
scale	-0.0115360	0.0232815

Table 6: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Gompertz distribution.

	x
shape	0.0389625
rate	-4.8682858

	x
shape	0.0196805
rate	-4.7223446

Table 7: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Gompertz distribution.

	shape	rate
shape	0.0006747	-0.0056127
rate	-0.0056127	0.0636371

	shape	rate
shape	0.0000977	-0.0013452
rate	-0.0013452	0.0323508

Table 8: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Log-normal distribution.

	x
meanlog	4.5379239
sdlog	0.5475716

	x
meanlog	4.2306127
sdlog	0.4415553

Table 9: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Log-normal distribution.

	meanlog	sdlog
meanlog	0.0696811	0.0247171
sdlog	0.0247171	0.0114622

	meanlog	sdlog
meanlog	0.0552019	0.0201740
sdlog	0.0201740	0.0100416

Table 10: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Log-logistic distribution.

	x
shape	0.2450486
scale	4.0866536

	x
shape	0.3083347
scale	3.9564286

Table 11: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Log-logistic distribution.

	shape	scale
shape	0.0144929	-0.0213003
scale	-0.0213003	0.0434085

	shape	scale
shape	0.0110089	-0.0141002
scale	-0.0141002	0.0284142

Table 12: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Gen. Gamma distribution.

	x
mu	4.1703317
sigma	-0.3599072
Q	1.2031485

	x
mu	4.1841067
sigma	-0.3715486
Q	1.2355243

Table 13: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Gen. Gamma distribution.

	mu	sigma	Q
mu	0.1026807	0.2899736	-0.3589428
sigma	0.2899736	1.2303463	-1.6290913
Q	-0.3589428	-1.6290913	2.1837103

	mu	sigma	Q
mu	0.0214967	0.0038493	0.0109325
sigma	0.0038493	0.2125792	-0.3117525
Q	0.0109325	-0.3117525	0.4784056

3.2 Effective sample size plots

The below plots show the estimated effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the survival distributions under consideration. The vertical line shows the optimal value of alpha identified by the algorithm. Larger ESS values are better.

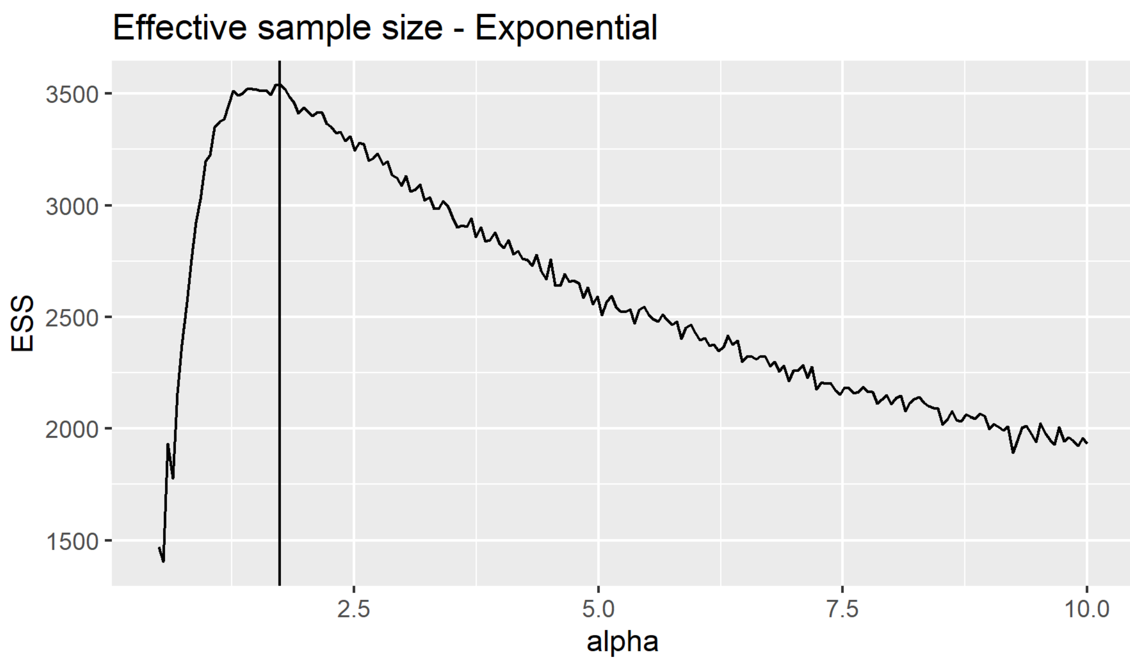

Figure 8: Effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the Exponential distribution. The vertical line shows the optimal value of alpha identified by the algorithm.

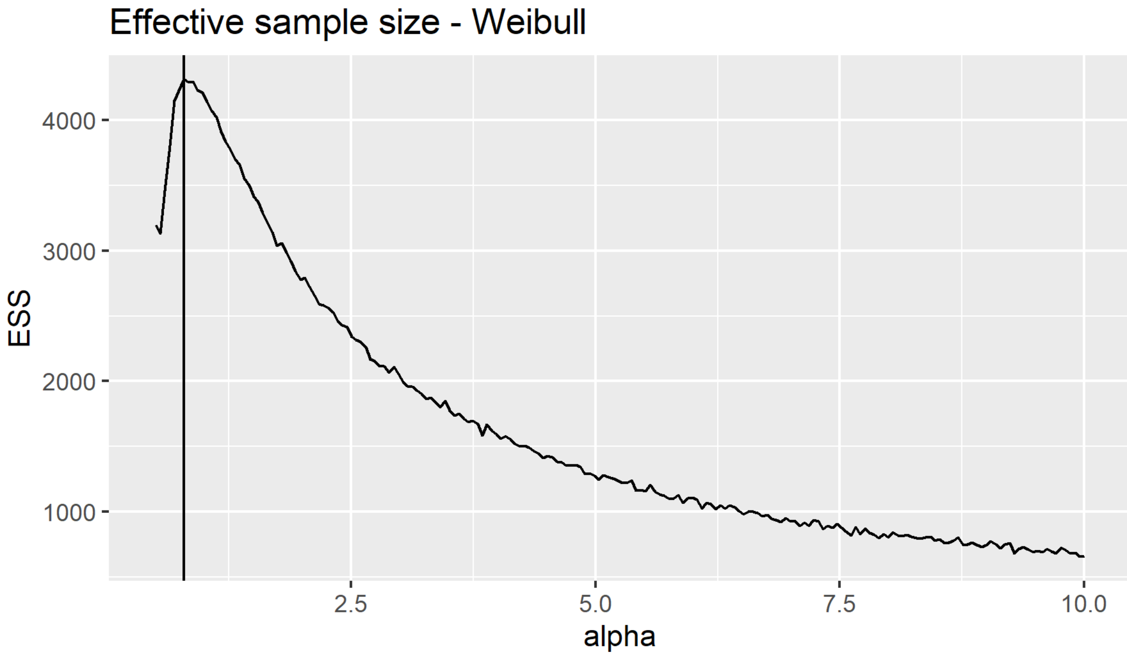

Figure 9: Effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the Weibull distribution. The vertical line shows the optimal value of alpha identified by the algorithm.

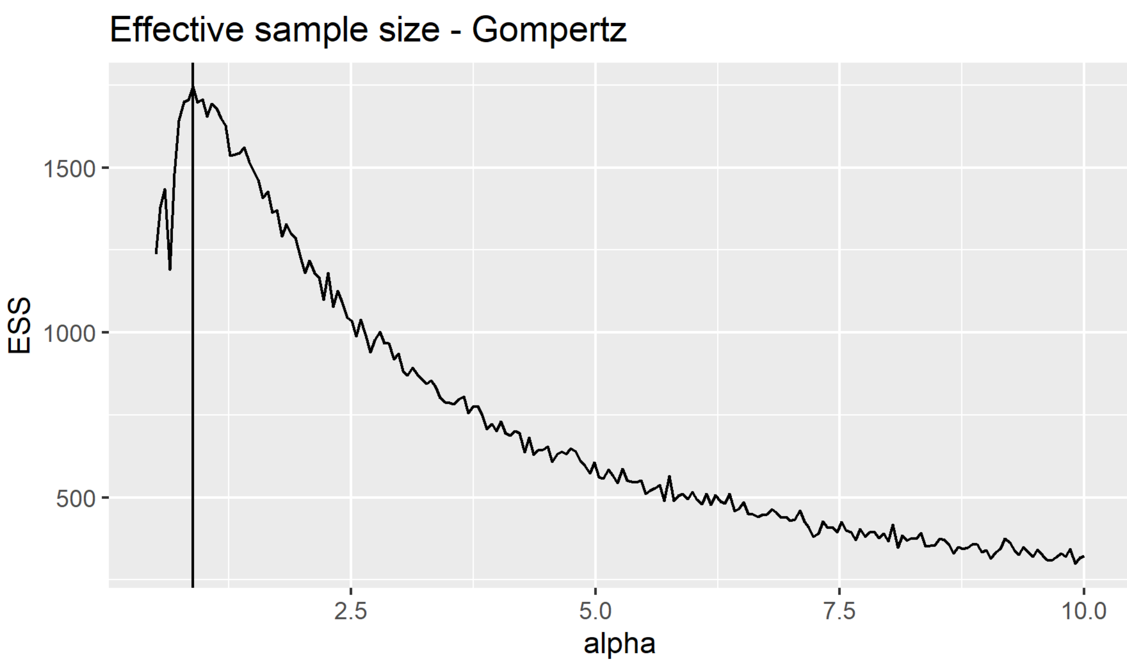

Figure 10: Effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the Gompertz distribution. The vertical line shows the optimal value of alpha identified by the algorithm.

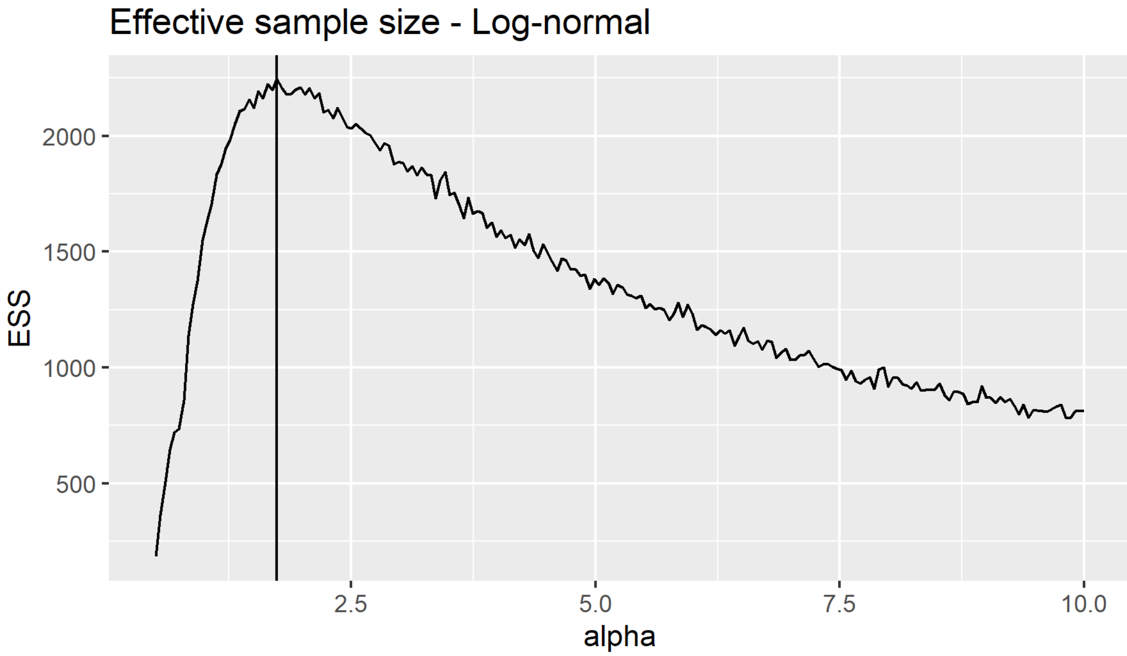

Figure 11: Effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the Log-normal distribution. The vertical line shows the optimal value of alpha identified by the algorithm.

Figure 12: Effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the Log-logistic distribution. The vertical line shows the optimal value of alpha identified by the algorithm.

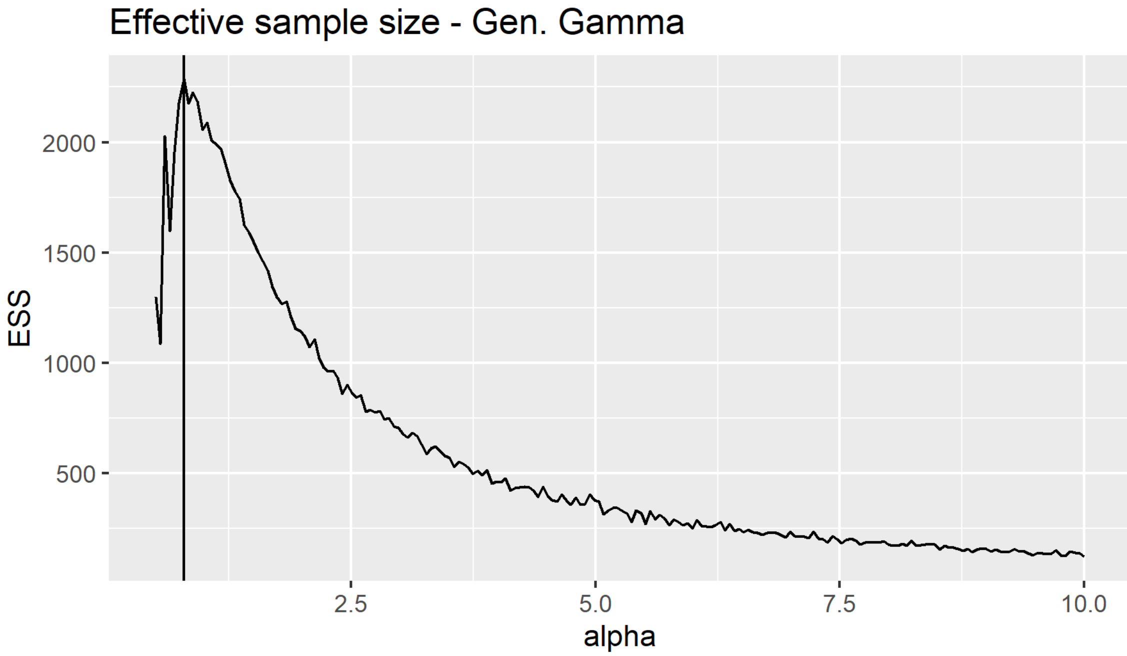

Figure 13: Effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the Gen. Gamma distribution. The vertical line shows the optimal value of alpha identified by the algorithm.

3.3 Distribution comparison plots

The below plots show parameter distributions and scatter plot estimates obtained using trial data only and by combining trial data with external information for the survival distributions under consideration.

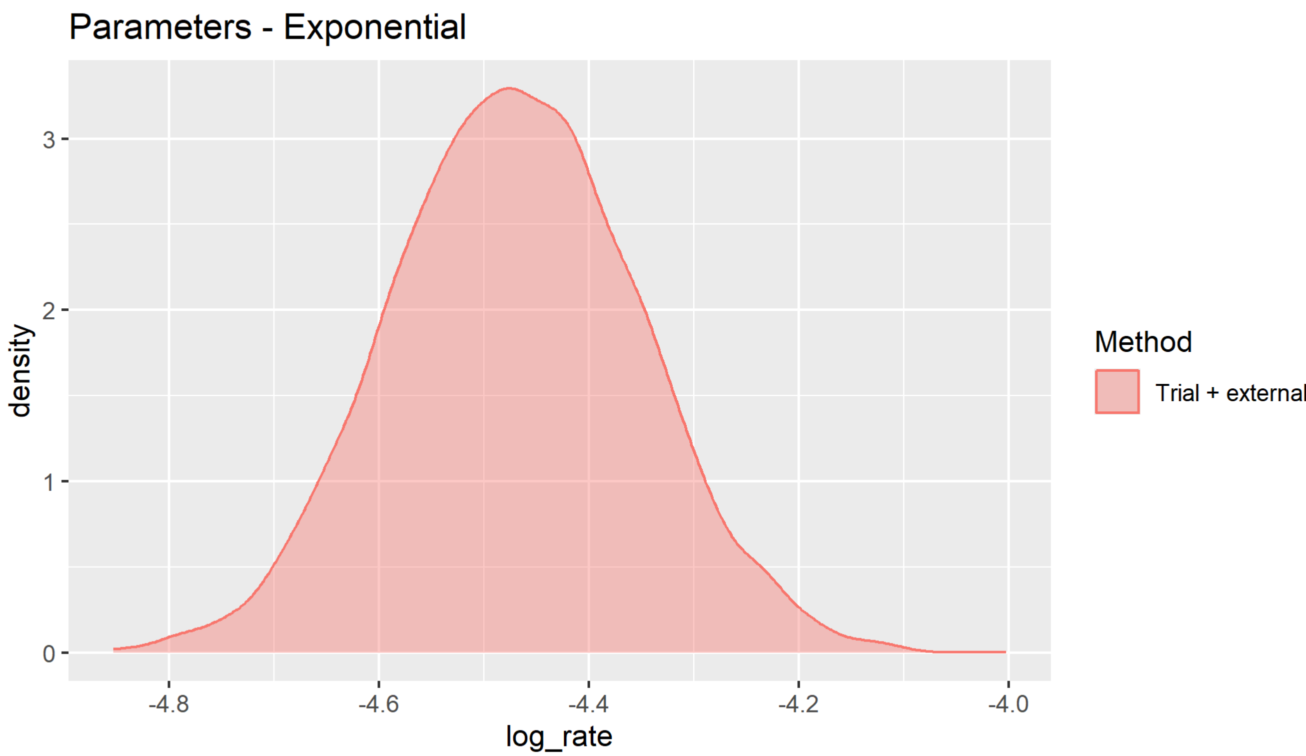

Figure 14: Survival curve parameter distributions and covariance estimates obtained using trial data only and combining trial data with external information, Exponential distribution.

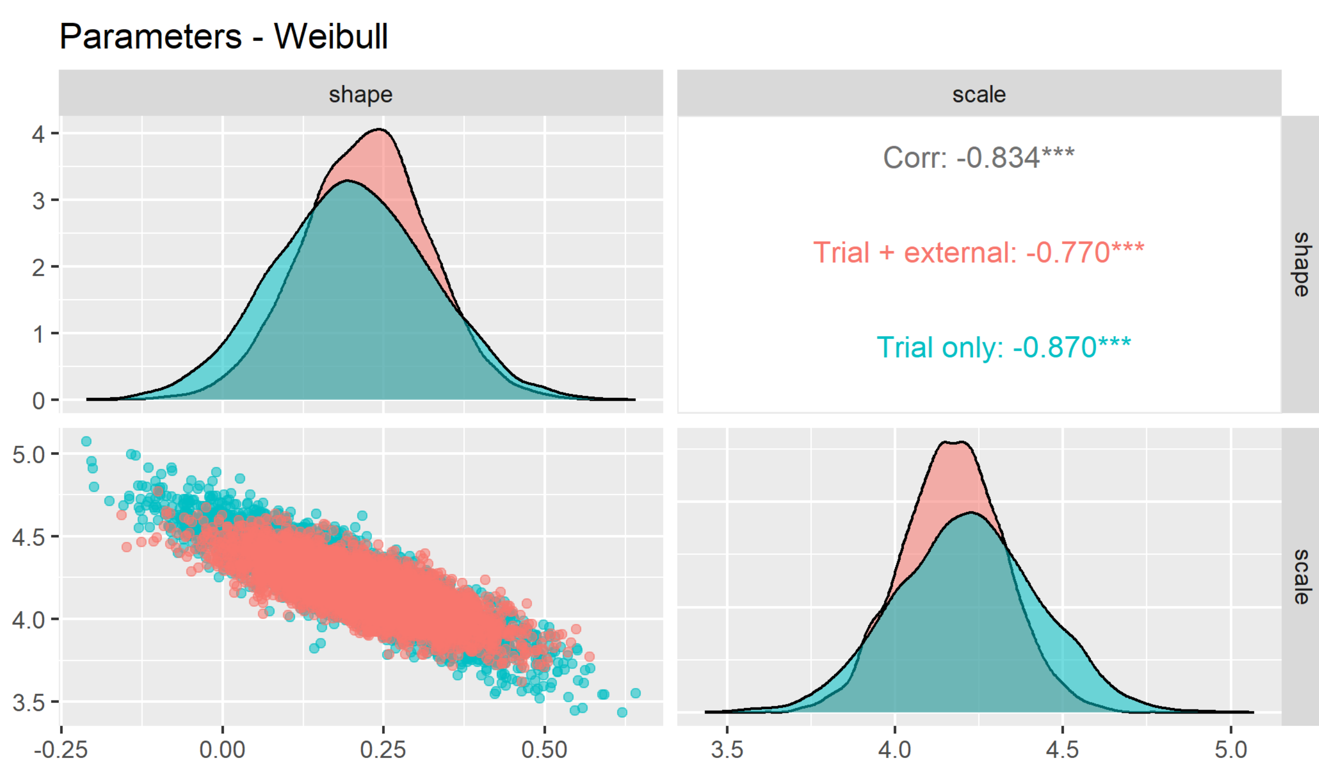

Figure 15: Survival curve parameter distributions and covariance estimates obtained using trial data only and combining trial data with external information, Weibull distribution.

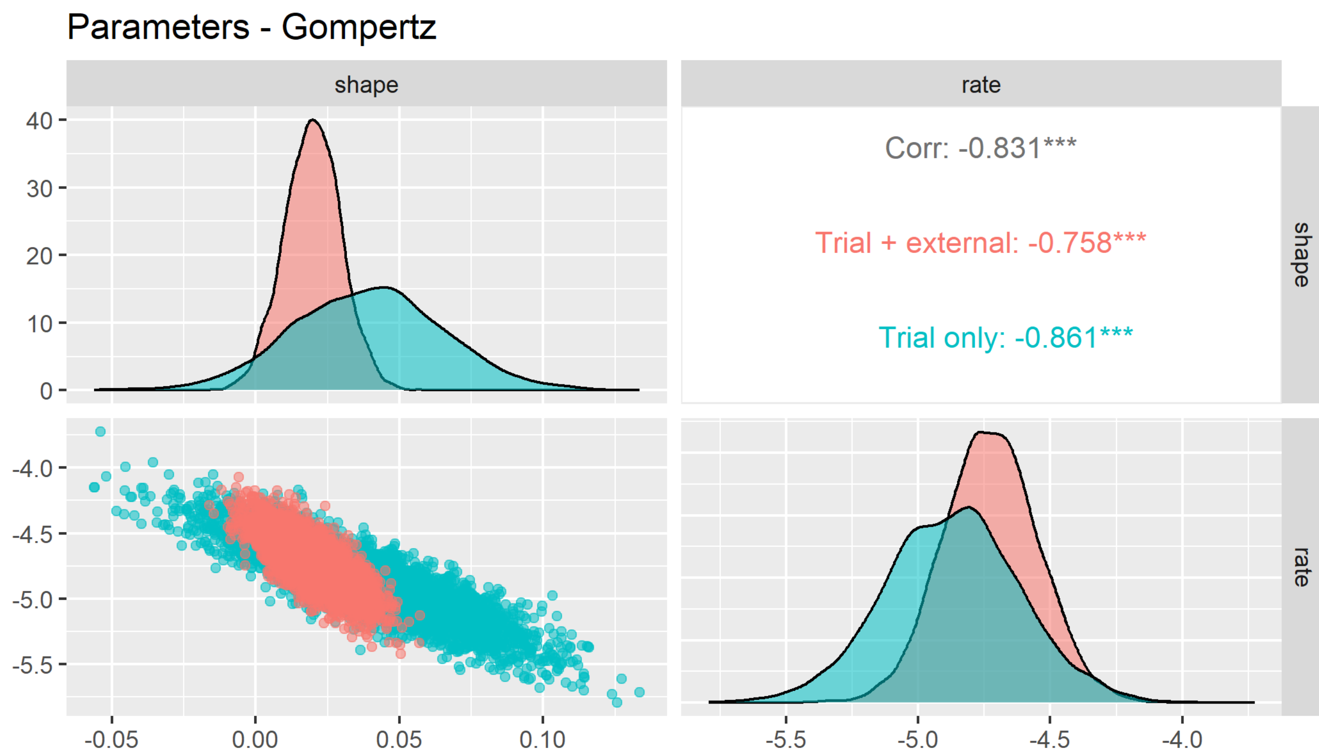

Figure 16: Survival curve parameter distributions and covariance estimates obtained using trial data only and combining trial data with external information, Gompertz distribution.

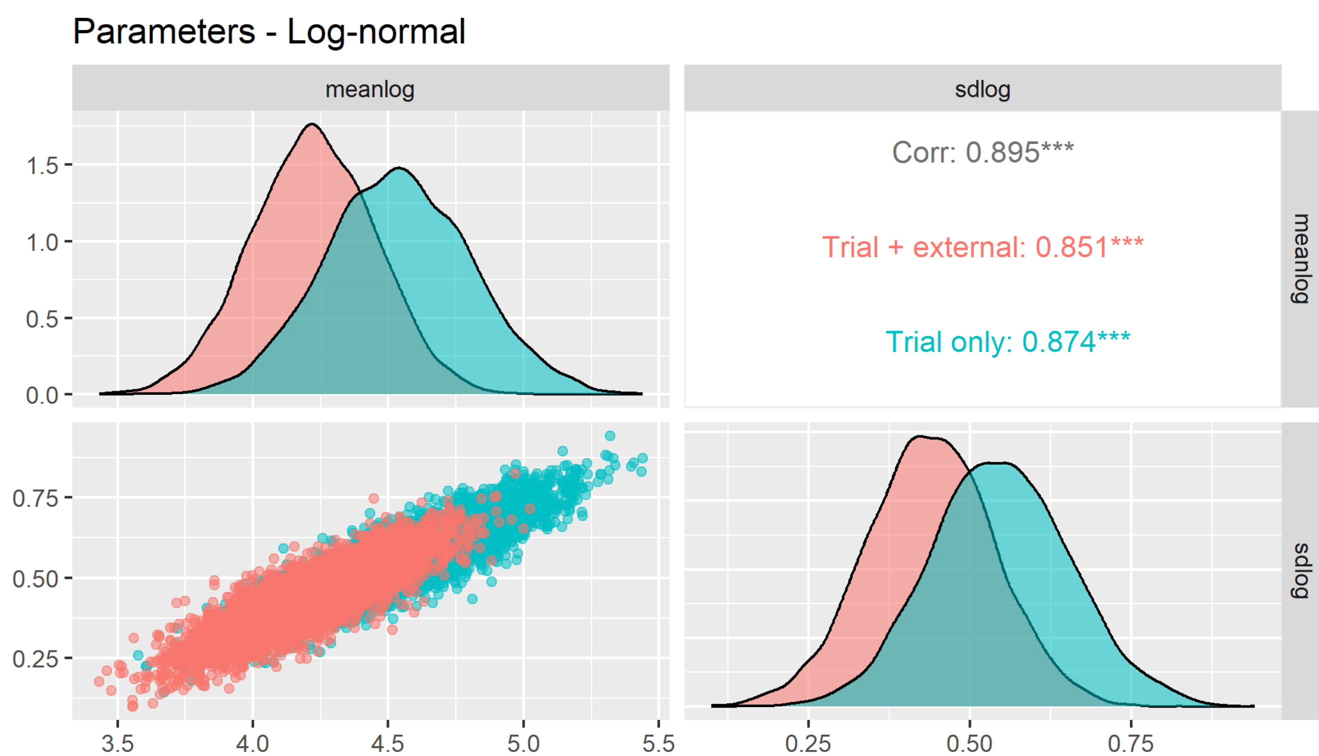

Figure 17: Survival curve parameter distributions and covariance estimates obtained using trial data only and combining trial data with external information, Log-normal distribution.

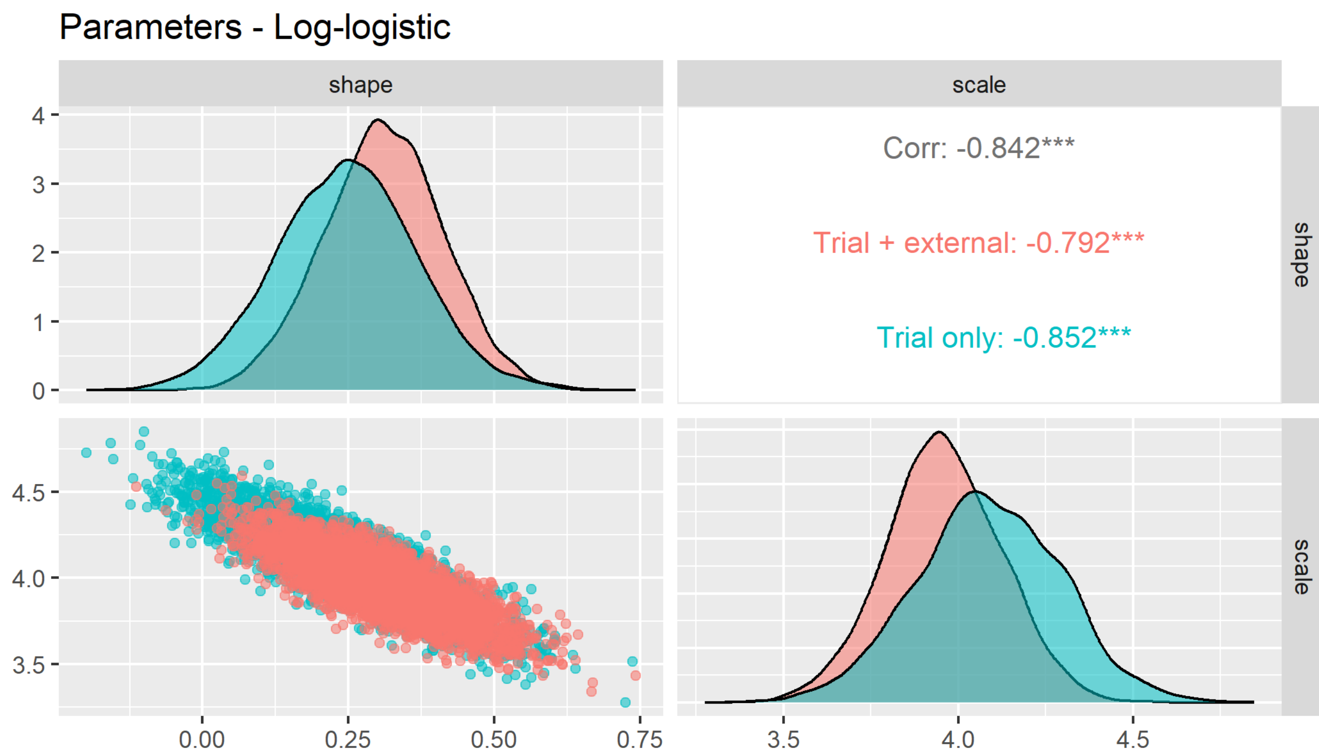

Figure 18: Survival curve parameter distributions and covariance estimates obtained using trial data only and combining trial data with external information, Log-logistic distribution.

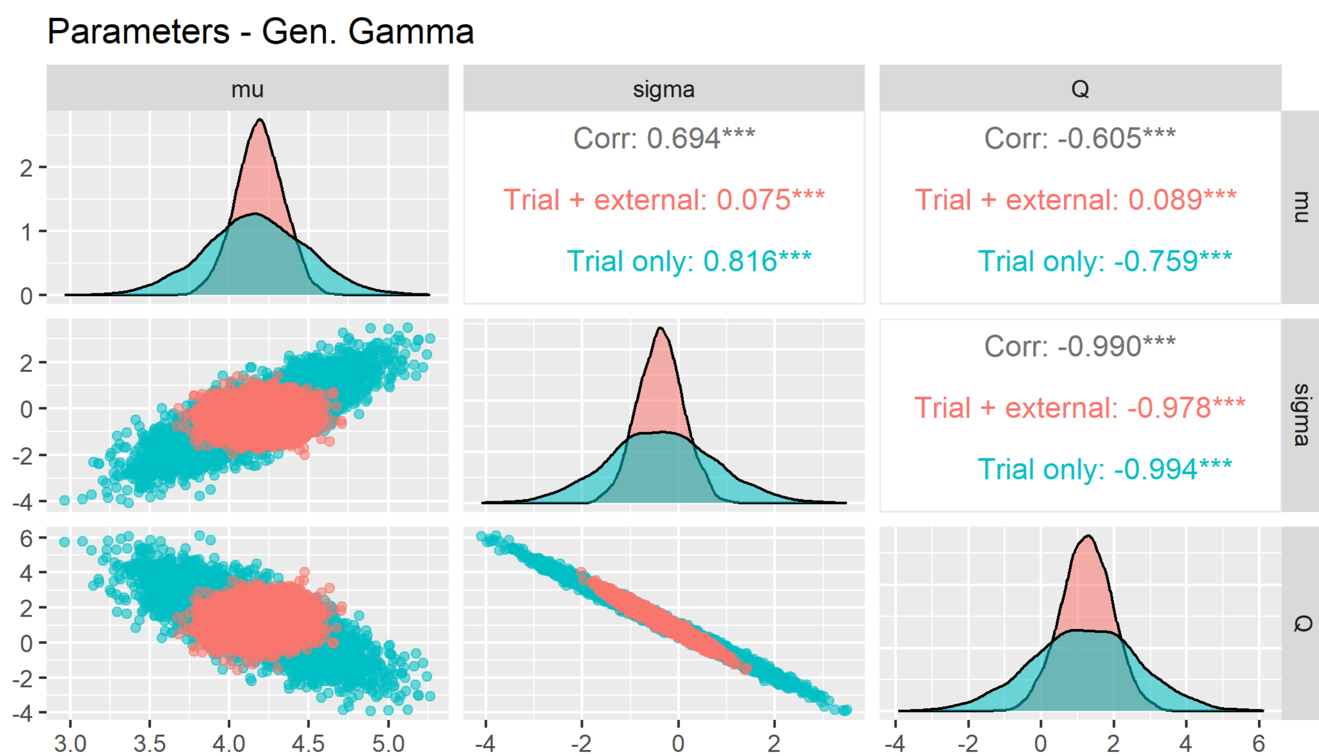

Figure 19: Survival curve parameter distributions and covariance estimates obtained using trial data only and combining trial data with external information, Gen. Gamma distribution.

3.4 Parameter variability

Table 14 shows 'Generalised variance,' i.e., determinants of variance-covariance matrices, of the parameter estimates obtained using trial data only (MLE) and combining trial data withe external information (IS). Formally, the generalised variance gives the area of the 95% highest density ellipse and can be interpreted as a 1-parameter measure of parameter uncertainty.

Table 14: Generalised variance of model parameter distributions. Comparison between parameter estimates obtained using trial data only (MLE) and combining trial data with external information (IS).

Distribution	Distance	MLE.Variance	IS.Variance	Var..Ratio
exponential	0.0957855	0.0169492	0.0138444	0.8168192
weibull	0.0399370	0.0001670	0.0000810	0.4852173
gompertz	0.1472096	0.0000114	0.0000014	0.1182030
lognormal	0.3250839	0.0001878	0.0001473	0.7846255
llogis	0.1447884	0.0001754	0.0001140	0.6498618
gengamma	0.0370603	0.0003562	0.0000382	0.1072668

4 Comparison with expertsurv output

In this section we compare the outputs obtained from our sansIPD (importance sampling) method to that obtained using the Cooney & White method implemented in the expertsurv package. For the gen. Gamma model estimated using the expertsurv package, suitable Monte Carlo performance could not be obtained and hence results are not reported.

4.1 Plots of survival curves over time

Figures 20 to 24 show comparisons of survival curves (median and 95% confidence/credible intervals) obtained from the sansIPD method described in the paper, with those obtained from the approach of Cooney & White implemented in the expertsurv package.

Survival curves for the exponential, Weibull, Gompertz and log-logistic models were almost identical between the two methods. For the log-normal model (Figure 23) there was a small but notable shift in the parameter distribution and corresponding survival estimate. The reason for this is not immediately clear. However, in both cases the impact was small and mostly evident in the tail of the survival distribution.

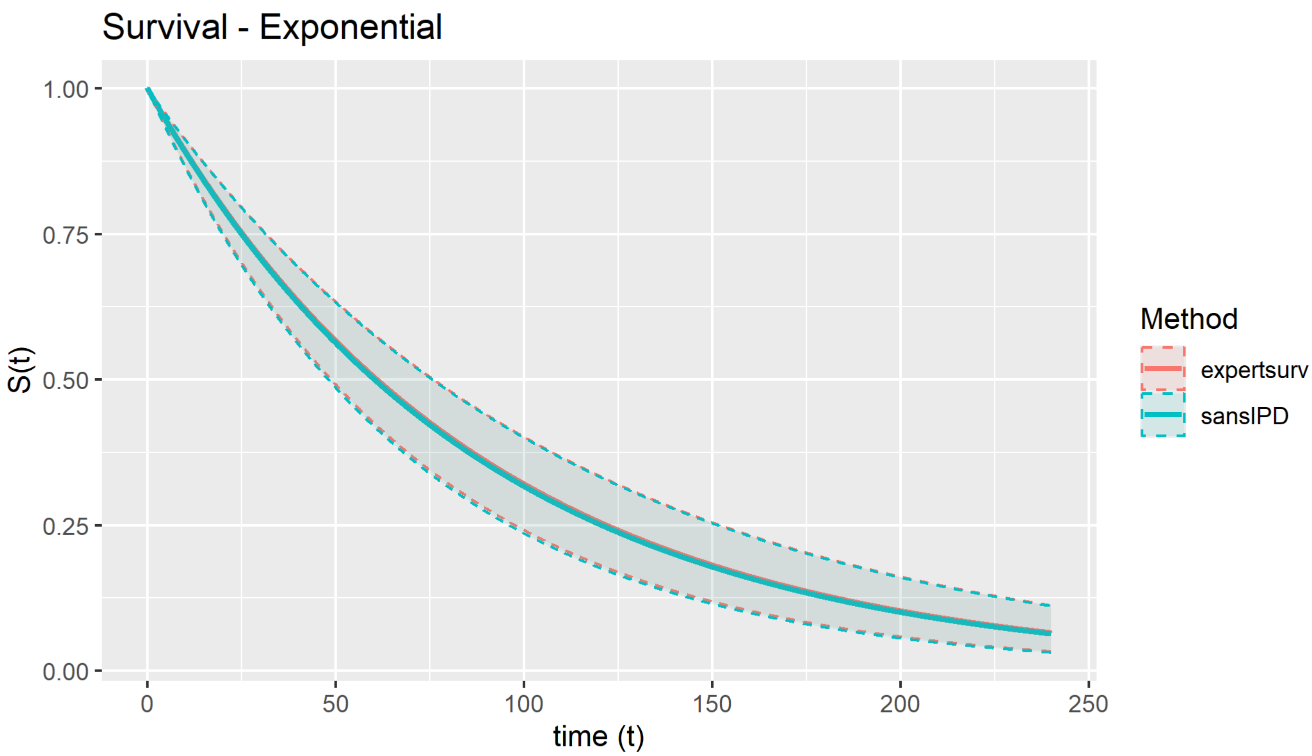

Figure 20: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the sansIPD and expertsurv methods, Exponential distribution

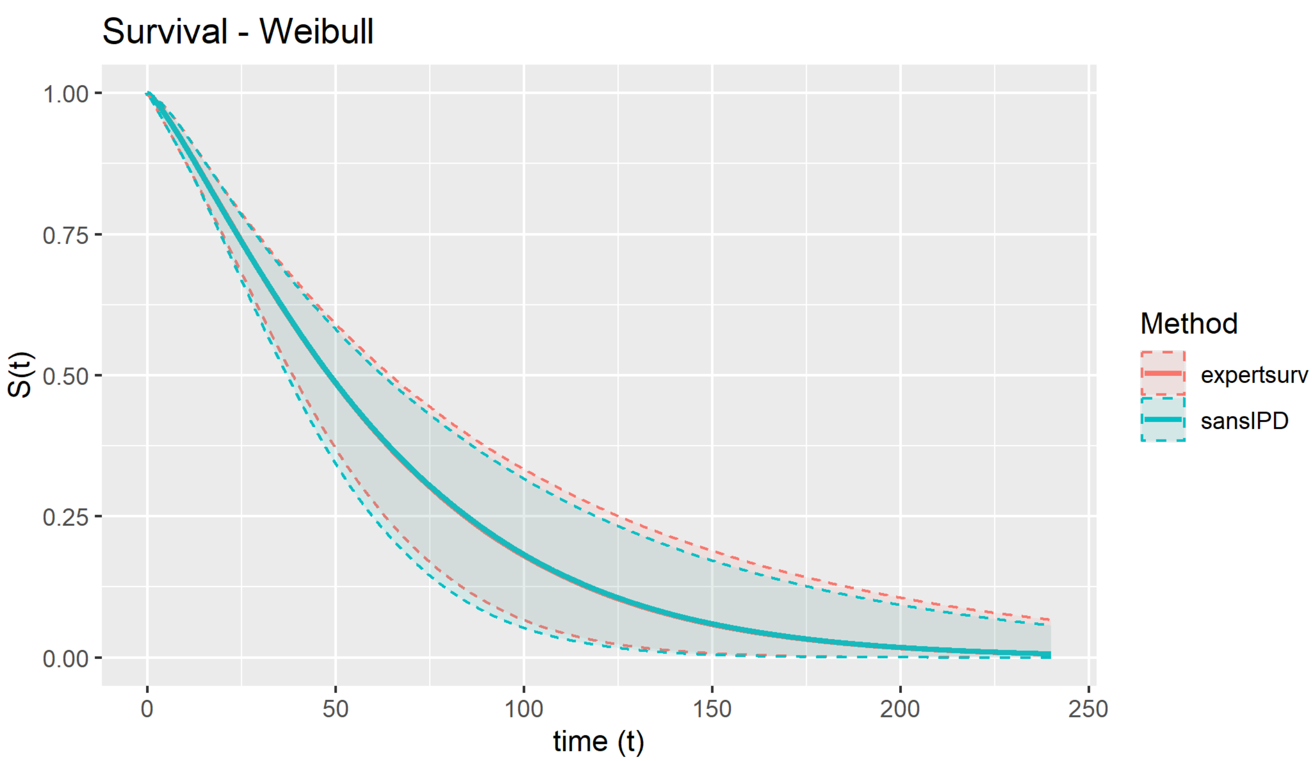

Figure 21: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the sansIPD and expertsurv methods, Weibull distribution

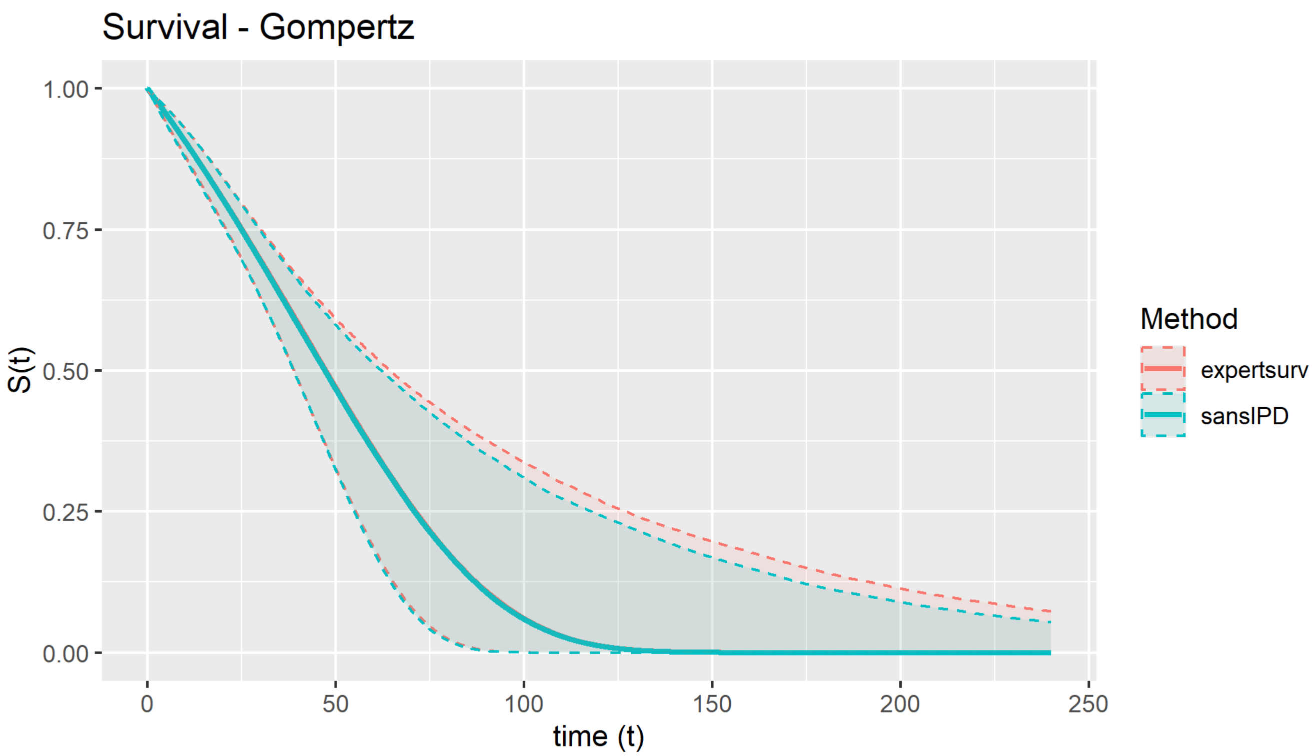

Figure 22: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the sansIPD and expertsurv methods, Gompertz distribution

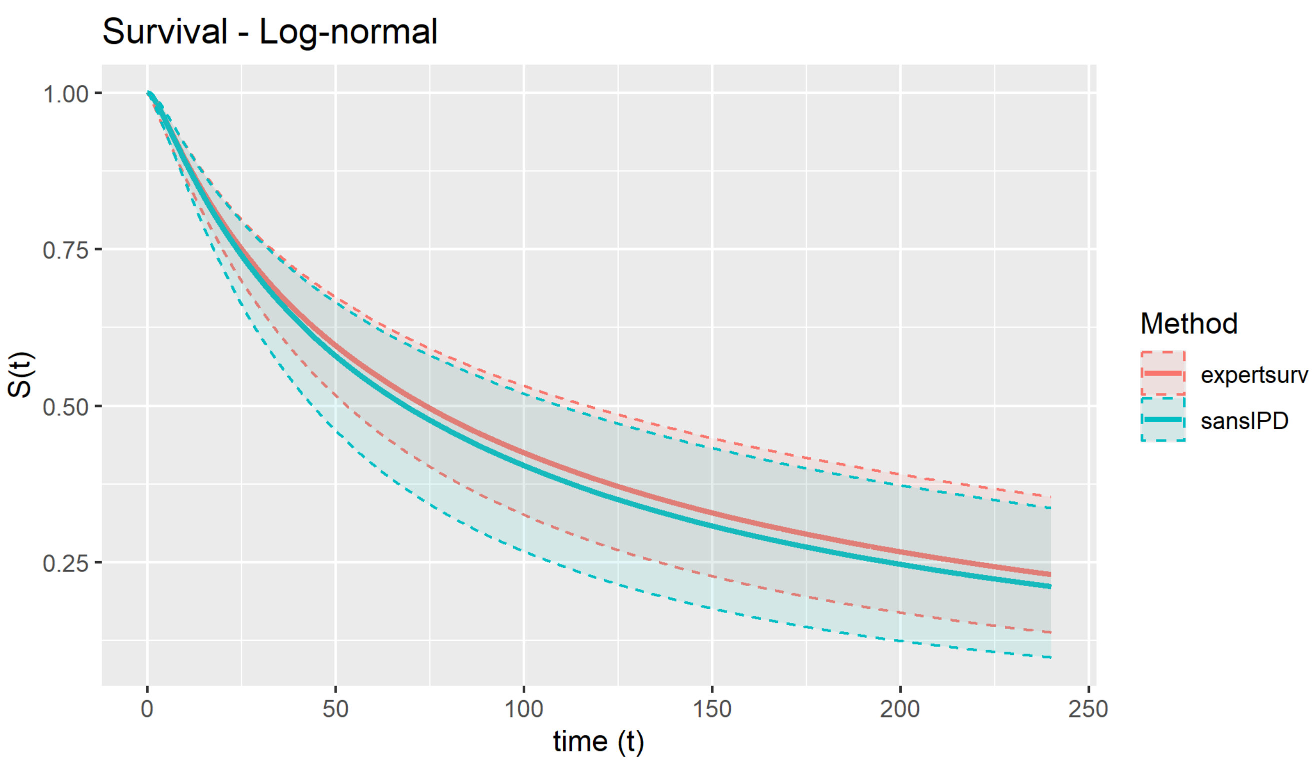

Figure 23: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the sansIPD and expertsurv methods, Log-normal distribution

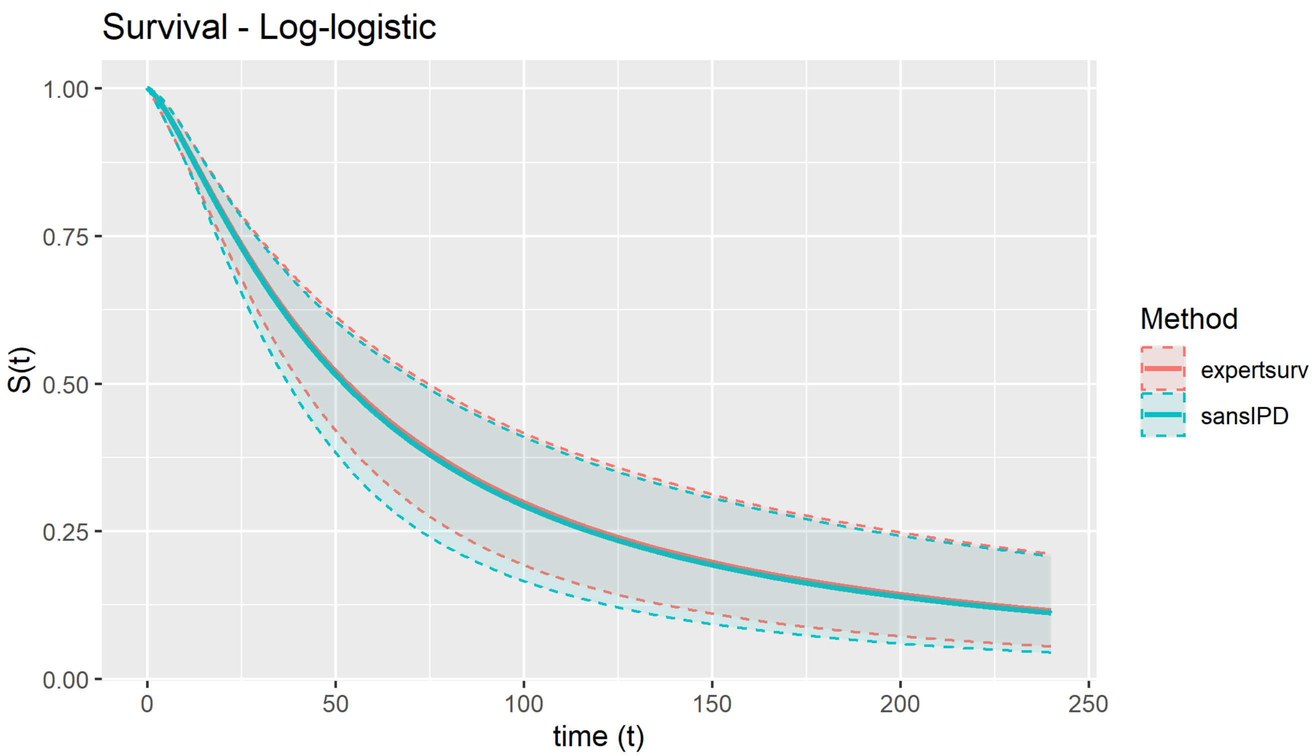

Figure 24: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the sansIPD and expertsurv methods, Log-logistic distribution

4.2 Comparisons of parameter distributions

Figures 25 to 29 compare survival curve parameter distributions obtained from the sansIPD method described in the paper, with those obtained from the approach of Cooney & White implemented in the expertsurv package. Again, some differences are evident between the two methods for the log-normal model (Figure 28).

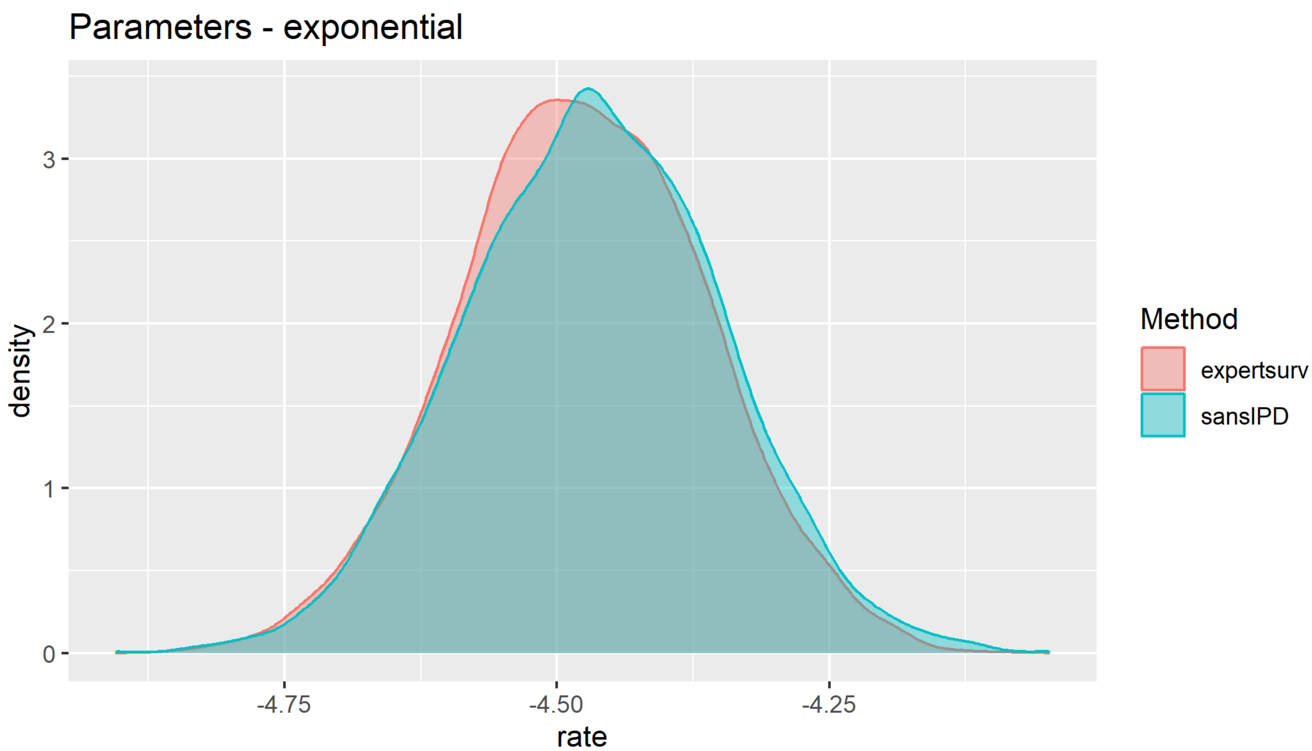

Figure 25: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the sansIPD and expertsurv methods, Exponential distribution

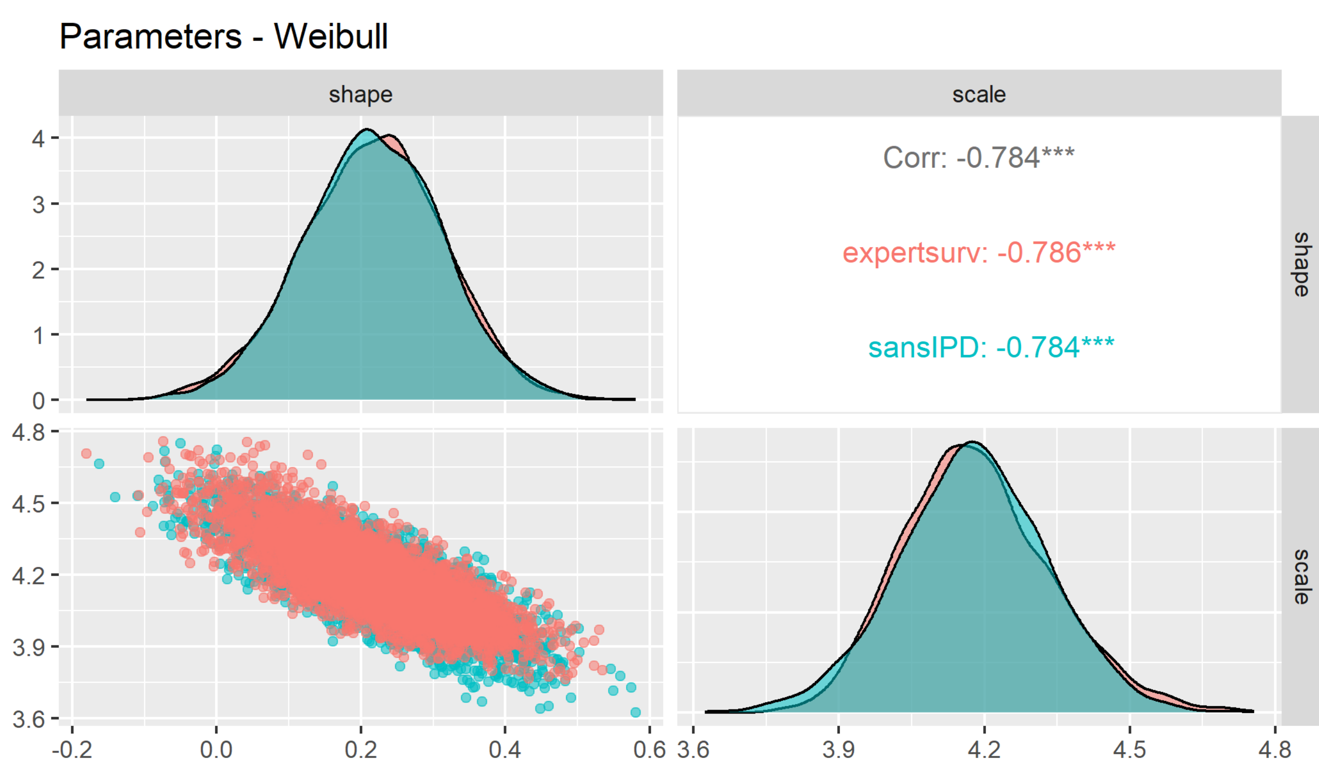

Figure 26: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the sansIPD and expertsurv methods, Weibull distribution

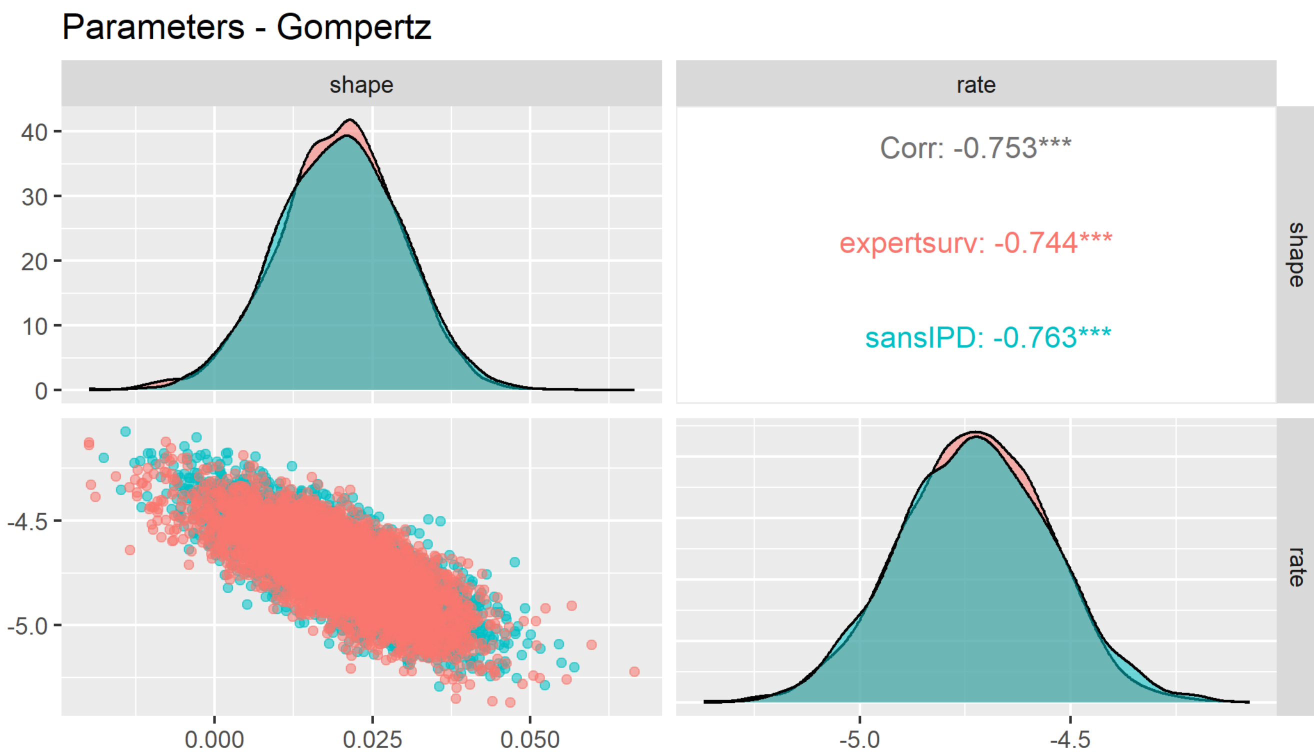

Figure 27: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the sansIPD and expertsurv methods, Gompertz distribution

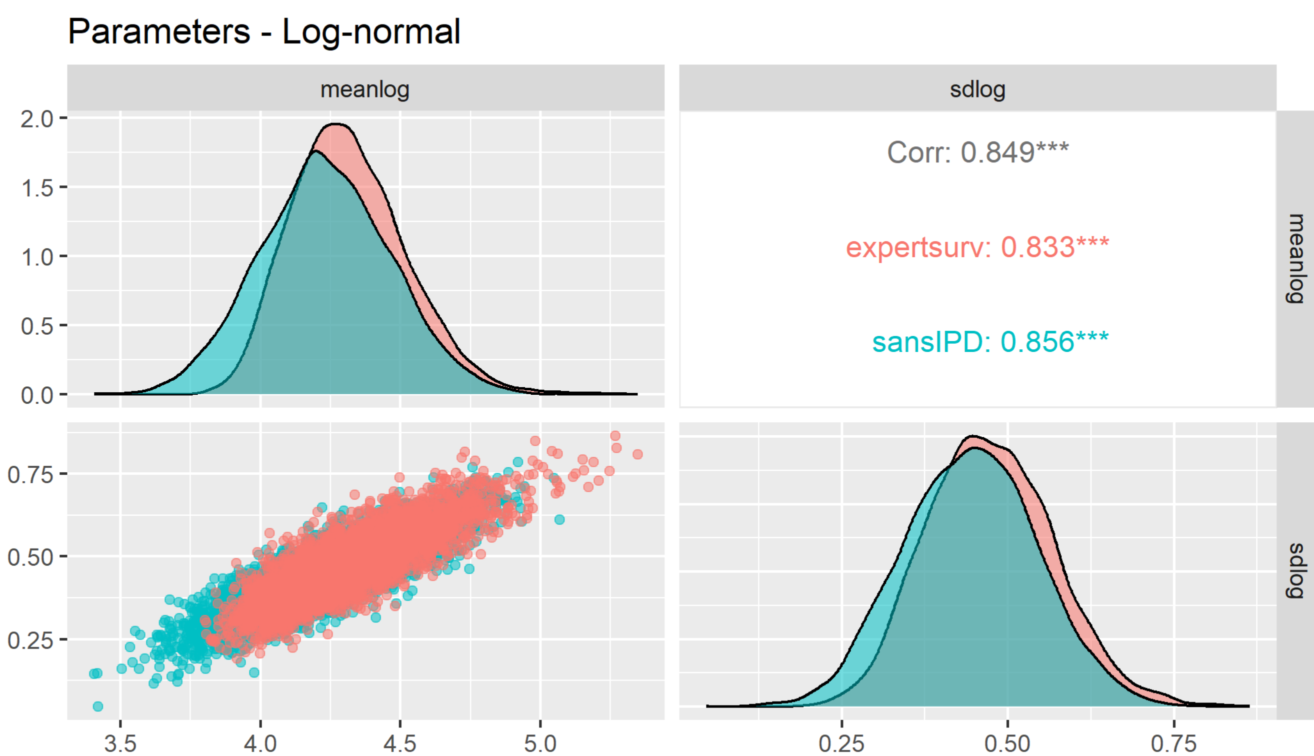

Figure 28: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the sansIPD and expertsurv methods, Log-normal distribution

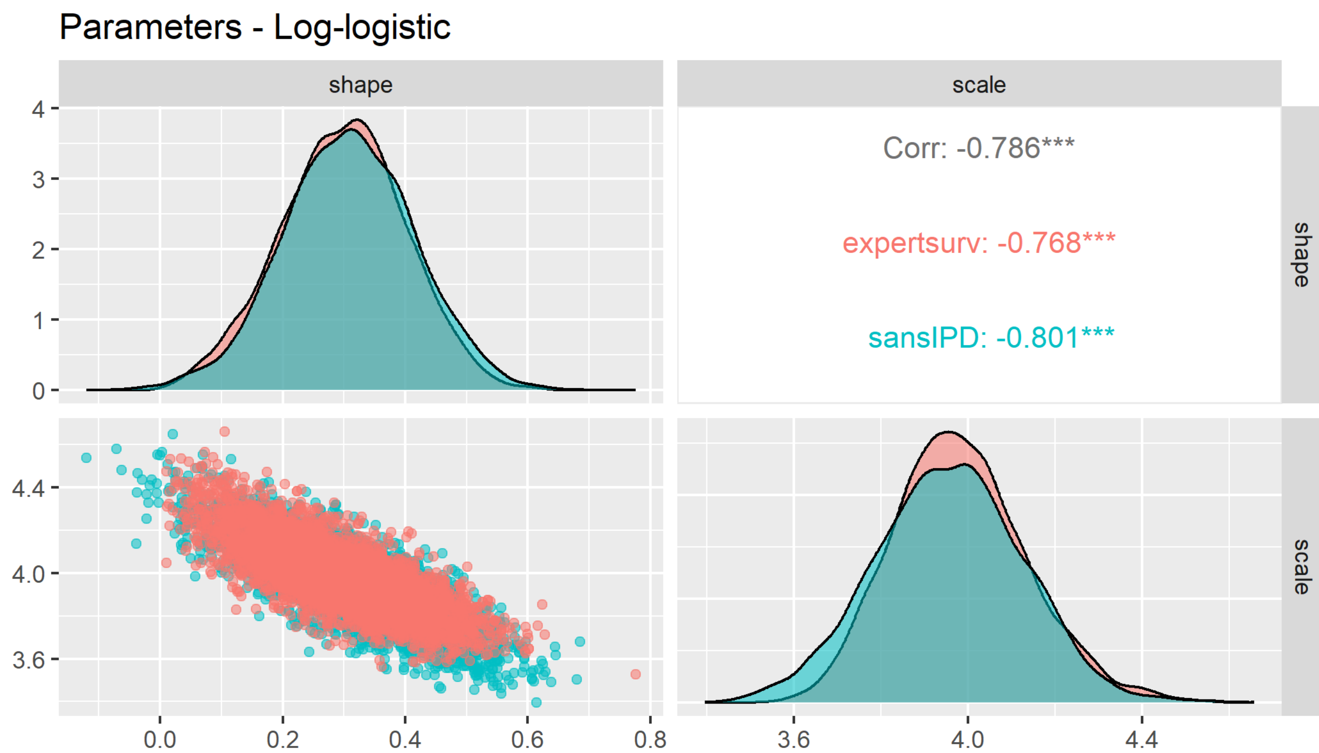

Figure 29: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the sansIPD and expertsurv methods, Log-logistic distribution

4.3 Comparisons of Mean OS

Table 15 and Figure 30 compare probabilistic estimates of mean OS between the two methods. In these examples survival is capped at 100 years since some curves plateau.

Table 15: Comparison of probabilistic total area under the curve (mean lifetime OS) estimates, obtained from combining trial data with external information using the sansIPD method described in the paper, and with the method of Cooney and White implemented in the expertsurv package. Values shown are probabilstic mean and 95 percent confidence/credible intervals.

Distribution	expertsurv	sansIPD	Mean.Diff	Var.Ratio
Exponential	88.48 (70.48, 109.73)	87.69 (69.33, 109.25)	-0.79	1.07
Gompertz	55.85 (39.56, 92.84)	53.66 (39.41, 84.54)	-2.20	0.41
Log-logistic	123.17 (77.65, 189.59)	119.55 (69.74, 186.52)	-3.63	1.10
Log-normal	202.54 (126.08, 311.84)	184.78 (99.17, 293.62)	-17.76	1.07
Weibull	62.43 (45.73, 90.16)	61.86 (43.28, 85.75)	-0.57	0.91

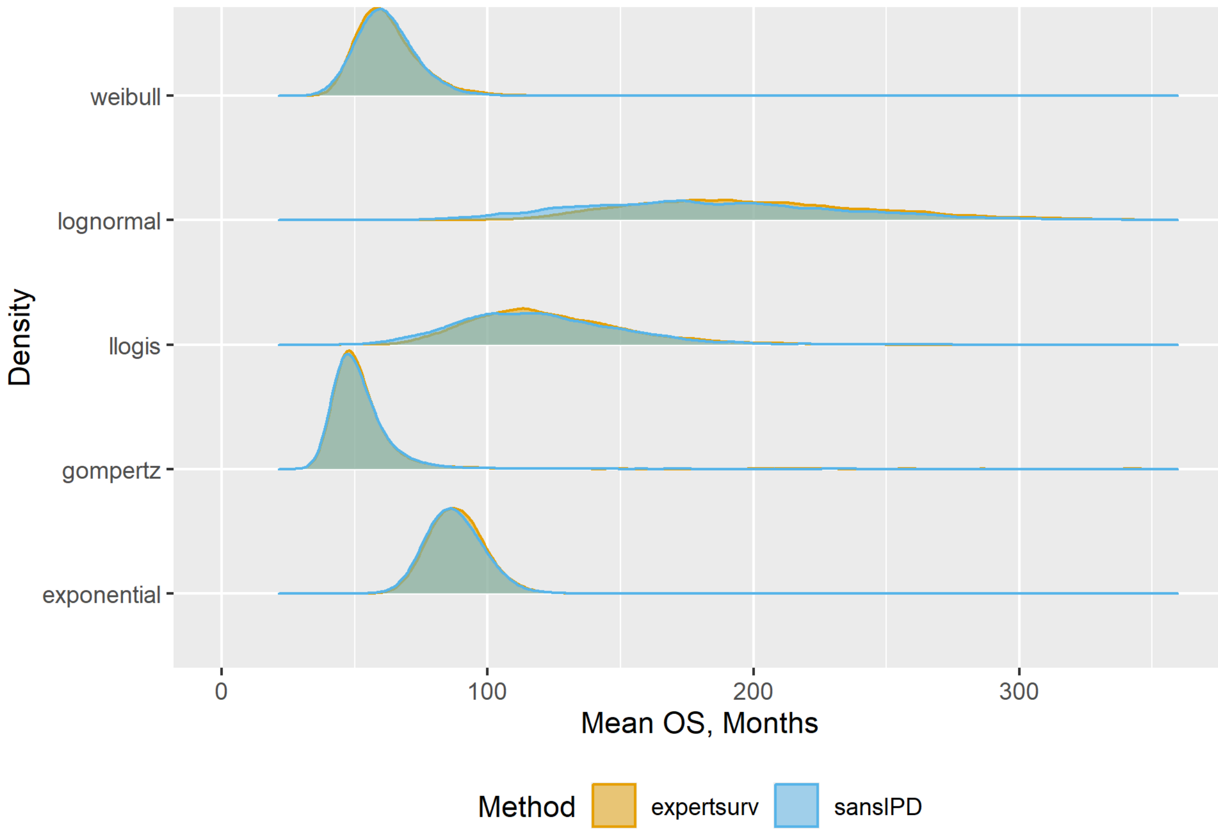

Figure 30: Density plots comparing probabilistic total area under the curve (mean OS) distributions obtained from combining trial data with external information using the sansIPD method described in the paper, with those obtained from the method of Cooney and White implemented in the expertsurv package.

4.4 Comparisons of 5-year OS probabilities

Table 16 and Figure 31 compare probabilistic estimates of 5-year OS obtained from the two methods.

Table 16: Comparison of 5-year OS estimates, obtained from combining trial data with external information using the sansIPD method described in the paper, with the method of Cooney and White implemented in the expertsurv package.

Distribution	Output	mean	median	sd	lwr.95	upr.95
Exponential	sansIPD	50.14%	50.20%	4.05%	42.09%	57.74%
Exponential	expertsurv	50.47%	50.56%	3.87%	42.68%	57.88%
Gompertz	sansIPD	35.48%	35.81%	8.73%	18.08%	51.27%
Gompertz	expertsurv	35.89%	36.01%	8.71%	18.84%	52.77%
Log-logistic	sansIPD	44.89%	45.33%	6.14%	31.37%	55.50%
Log-logistic	expertsurv	45.88%	45.95%	5.37%	35.19%	56.30%
Log-normal	sansIPD	52.98%	53.35%	5.71%	40.58%	62.78%
Log-normal	expertsurv	55.11%	55.20%	4.50%	46.42%	63.67%
Weibull	sansIPD	39.98%	40.57%	6.74%	24.86%	51.52%
Weibull	expertsurv	40.32%	40.41%	6.53%	27.64%	52.69%

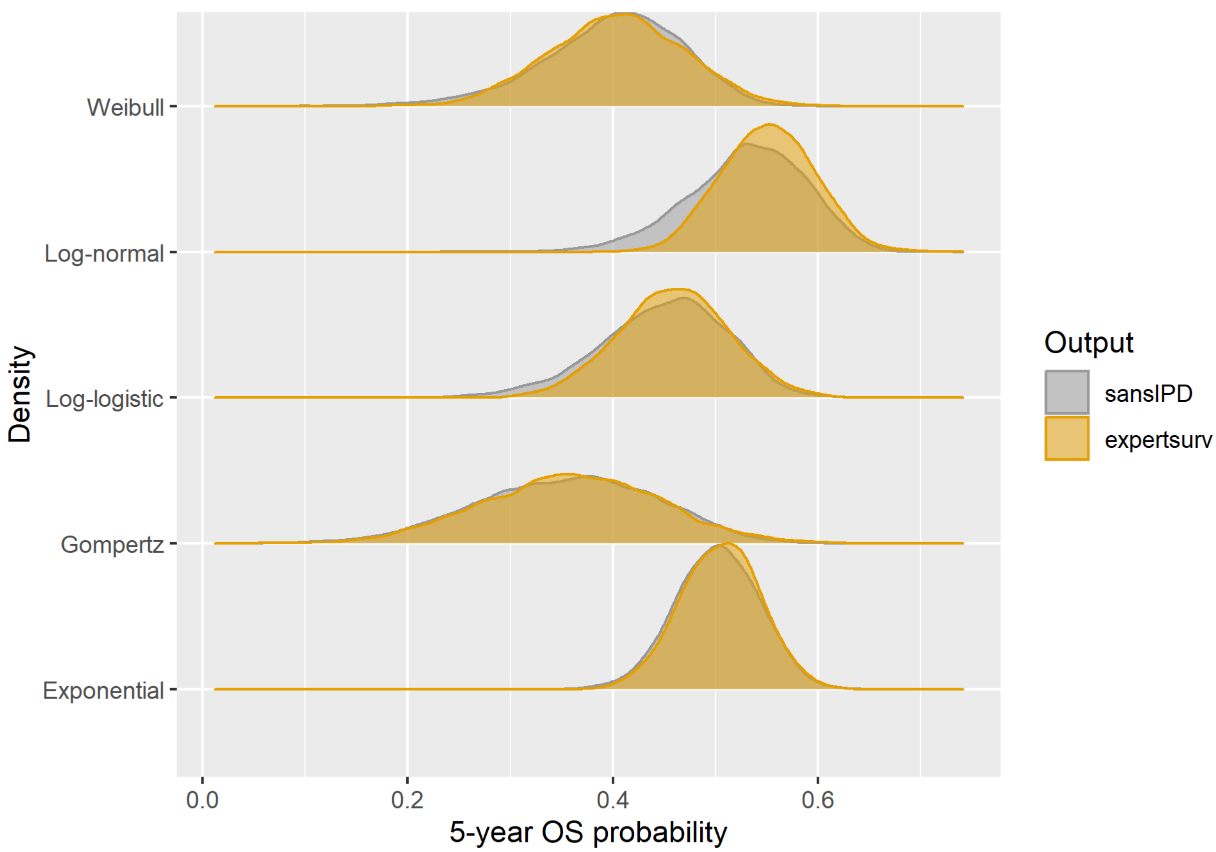

Figure 31: Density plots comparing 5-year OS estimates, obtained from combining trial data with external information using the sansIPD method described in the paper, with those obtained from the method of Cooney and White implemented in the expertsurv package.

5 Scenario analysis: alternative specifications of external information

5.1 Description of included scenarios

The base case analysis assumes a prior distribution for 5-year OS given by normal prior with central 95% density contained within the range (20% to 55%); this distribution has mean `0.375` and variance `0.0079719`. We consider five additional scenarios:

- Scenario 1: Alternative specification of base case: Beta distribution with same mean and variance
- Scenario 2: Base case variance halved ("more confident"), mean unchanged
- Scenario 3: Base case variance doubled ("less confident"), mean unchanged
- Scenario 4: Base case mean decreased by 20%, variance unchanged, Beta distribution ("wrong with same confidence").
- Scenario 5: Base case mean decreased by 20%, variance halved, Beta distribution ("wrong and more confident")
- Scenario 6: Base case mean decreased by 20%, variance doubled, Beta distribution ("wrong and less confident")

Table 17: Summary of external information used in the base case and scenario analysis

Scenario	Description	Distribution	Mean	Variance	Prior.95CrI
Base case	NA	Normal(0.375,0.0893^2)	0.375	0.00797	(0.2000 to 0.550)
Scenario 1	Alternative specification of base case using Beta distribution	Beta(10.650,17.750)	0.375	0.00797	(0.2099 to 0.557)
Scenario 2	More confident	Normal(0.375,0.0631^2)	0.375	0.00399	(0.2513 to 0.499)
Scenario 3	Less confident	Normal(0.375,0.1263^2)	0.375	0.01594	(0.1275 to 0.622)
Scenario 4	Wrong with same confidence	Beta(2.994,14.116)	0.175	0.00797	(0.0400 to 0.381)
Scenario 5	Wrong and more confident	Beta(6.164,29.057)	0.175	0.00399	(0.0703 to 0.315)
Scenario 6	Wrong and less confident	Beta(1.410, 6.646)	0.175	0.01594	(0.0129 to 0.483)

5.2 Results of Scenario analyses:

For all scenarios, we plot posterior mean survival curve estimates, and compare mean lifetime OS and 5-year OS probabilities.

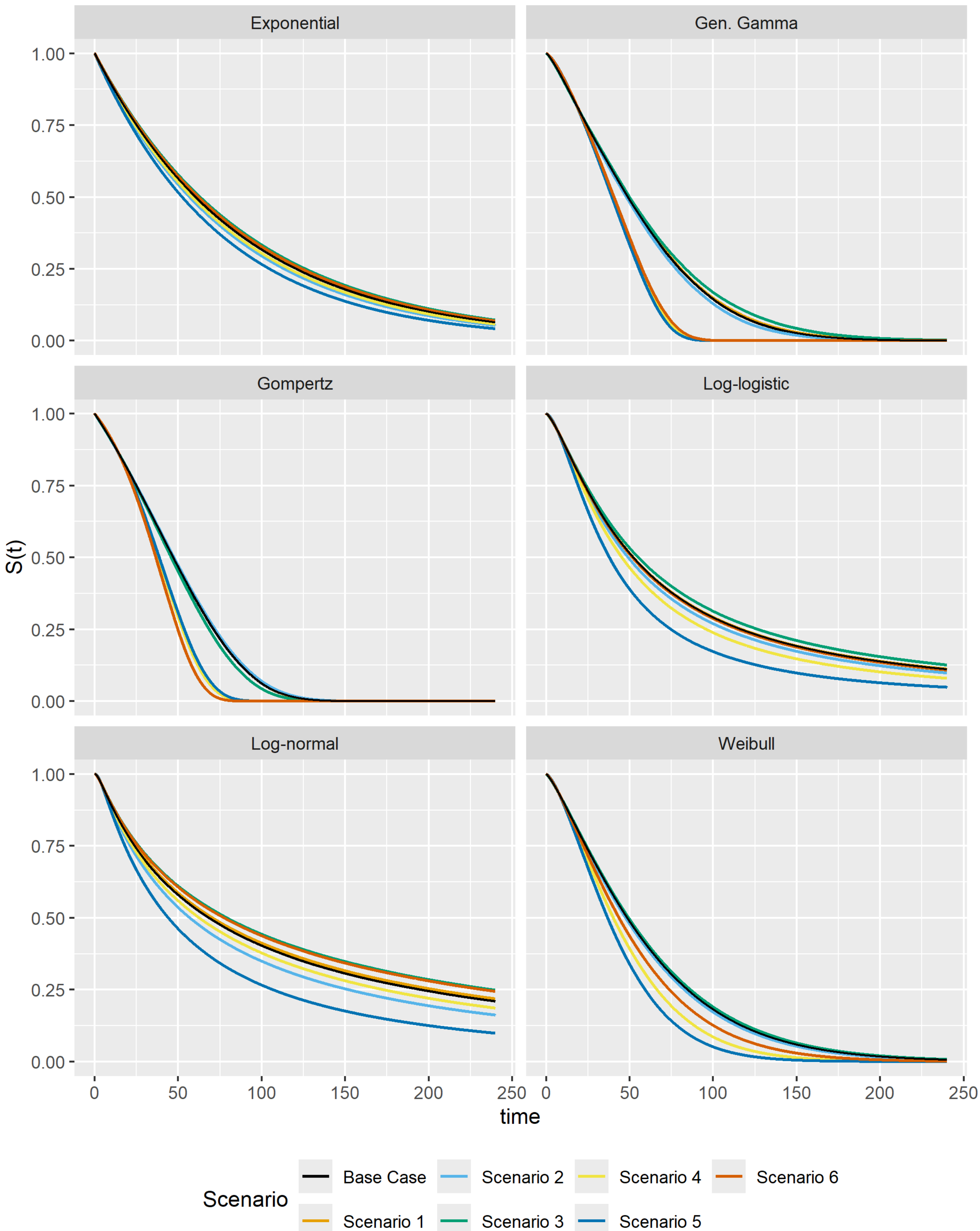

Figure 32: Survival curves (posterior median) for the base case and scenarios 1 through 6

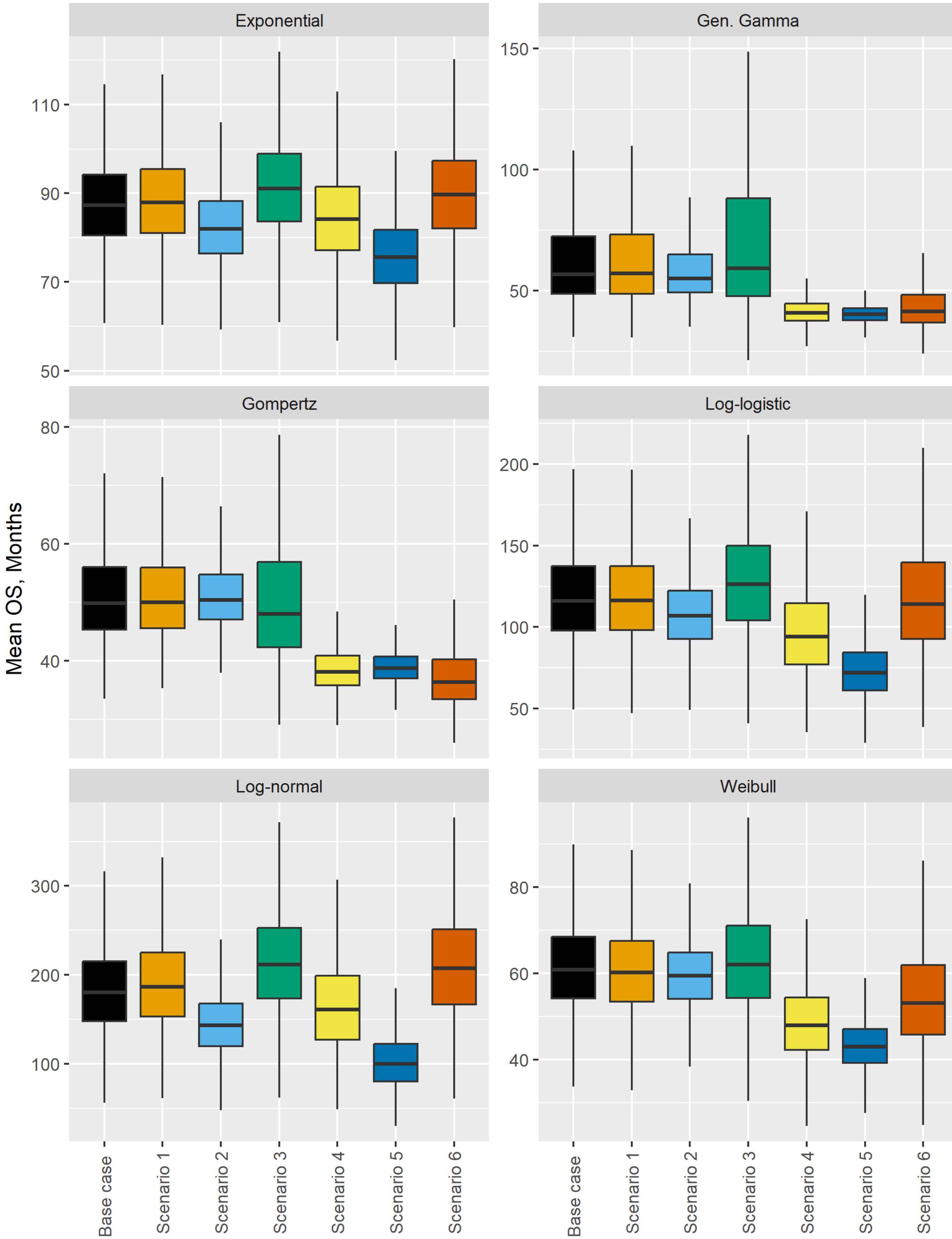

Figure 33: Boxplots of probabilistic simulations of mean OS in the Base Case and in Scenarios 1 through 5.

Table 18: Mean OS for Scenarios 1 through 5, including comparisons with base case (probabilistic mean difference and variance ratio)

					Scenario versus Base Case	
Distribution	Scenario	Mean	Median	CrI.95	Mean.Difference	Variance.Ratio
Exponential	Base case	87.74	87.27	(69.78, 109.28)	NA	NA
Exponential	Scenario 1	88.58	87.96	(69.44, 110.55)	0.84	1.06
Exponential	Scenario 2	82.55	81.92	(66.72, 101.16)	-5.18	0.73
Exponential	Scenario 3	91.71	91.10	(71.74, 115.47)	3.97	1.19
Exponential	Scenario 4	84.80	84.17	(65.28, 107.95)	-2.94	1.08
Exponential	Scenario 5	76.02	75.55	(59.77, 94.90)	-11.72	0.76
Exponential	Scenario 6	90.30	89.66	(69.29, 115.76)	2.56	1.28
Gen. Gamma	Base case	71.11	56.84	(38.84, 216.27)	NA	NA
Gen. Gamma	Scenario 1	71.55	57.09	(38.81, 211.96)	0.44	0.99
Gen. Gamma	Scenario 2	63.17	55.00	(41.63, 144.12)	-7.94	0.38
Gen. Gamma	Scenario 3	88.67	59.29	(36.17, 350.67)	17.56	2.87
Gen. Gamma	Scenario 4	41.98	40.85	(32.26, 58.69)	-29.14	0.03
Gen. Gamma	Scenario 5	40.80	40.25	(34.08, 51.10)	-30.31	0.01
Gen. Gamma	Scenario 6	45.67	41.55	(29.68, 86.64)	-25.45	0.18
Gompertz	Base case	53.89	49.80	(39.24, 85.24)	NA	NA
Gompertz	Scenario 1	53.99	50.00	(39.65, 85.91)	0.10	0.95
Gompertz	Scenario 2	52.00	50.38	(41.98, 70.15)	-1.90	0.15
Gompertz	Scenario 3	60.49	48.03	(35.52, 208.96)	6.59	5.72
Gompertz	Scenario 4	38.65	38.10	(32.29, 47.83)	-15.24	0.03
Gompertz	Scenario 5	39.00	38.71	(34.31, 45.28)	-14.90	0.01
Gompertz	Scenario 6	37.77	36.34	(29.53, 52.75)	-16.13	0.13
Log-logistic	Base case	119.06	116.09	(71.11, 181.96)	NA	NA
Log-logistic	Scenario 1	119.12	116.37	(70.56, 183.86)	0.05	1.03
Log-logistic	Scenario 2	108.55	106.95	(71.00, 156.21)	-10.52	0.58
Log-logistic	Scenario 3	128.74	126.25	(72.27, 201.04)	9.67	1.36
Log-logistic	Scenario 4	97.94	94.25	(52.67, 162.94)	-21.12	1.00
Log-logistic	Scenario 5	74.10	71.85	(45.07, 116.21)	-44.96	0.40
Log-logistic	Scenario 6	118.18	114.00	(61.11, 199.20)	-0.89	1.51
Log-normal	Base case	183.80	179.96	(97.93, 295.22)	NA	NA
Log-normal	Scenario 1	190.32	186.35	(100.47, 302.01)	6.52	1.06
Log-normal	Scenario 2	145.94	143.12	(84.89, 226.86)	-37.86	0.52
Log-normal	Scenario 3	215.54	211.42	(115.04, 340.09)	31.74	1.32
Log-normal	Scenario 4	166.22	161.09	(78.12, 284.68)	-17.58	1.11
Log-normal	Scenario 5	103.60	99.77	(53.53, 176.61)	-80.20	0.40
Log-normal	Scenario 6	211.09	207.41	(106.90, 336.72)	27.29	1.45
Weibull	Base case	61.93	60.80	(44.18, 85.17)	NA	NA
Weibull	Scenario 1	61.30	60.19	(43.33, 86.41)	-0.63	1.04
Weibull	Scenario 2	59.92	59.40	(45.35, 77.94)	-2.01	0.58
Weibull	Scenario 3	63.50	62.06	(42.30, 92.77)	1.57	1.44
Weibull	Scenario 4	48.78	47.94	(33.25, 69.73)	-13.15	0.74
Weibull	Scenario 5	43.36	42.98	(33.15, 55.87)	-18.57	0.30
Weibull	Scenario 6	54.69	53.10	(34.40, 84.16)	-7.24	1.38

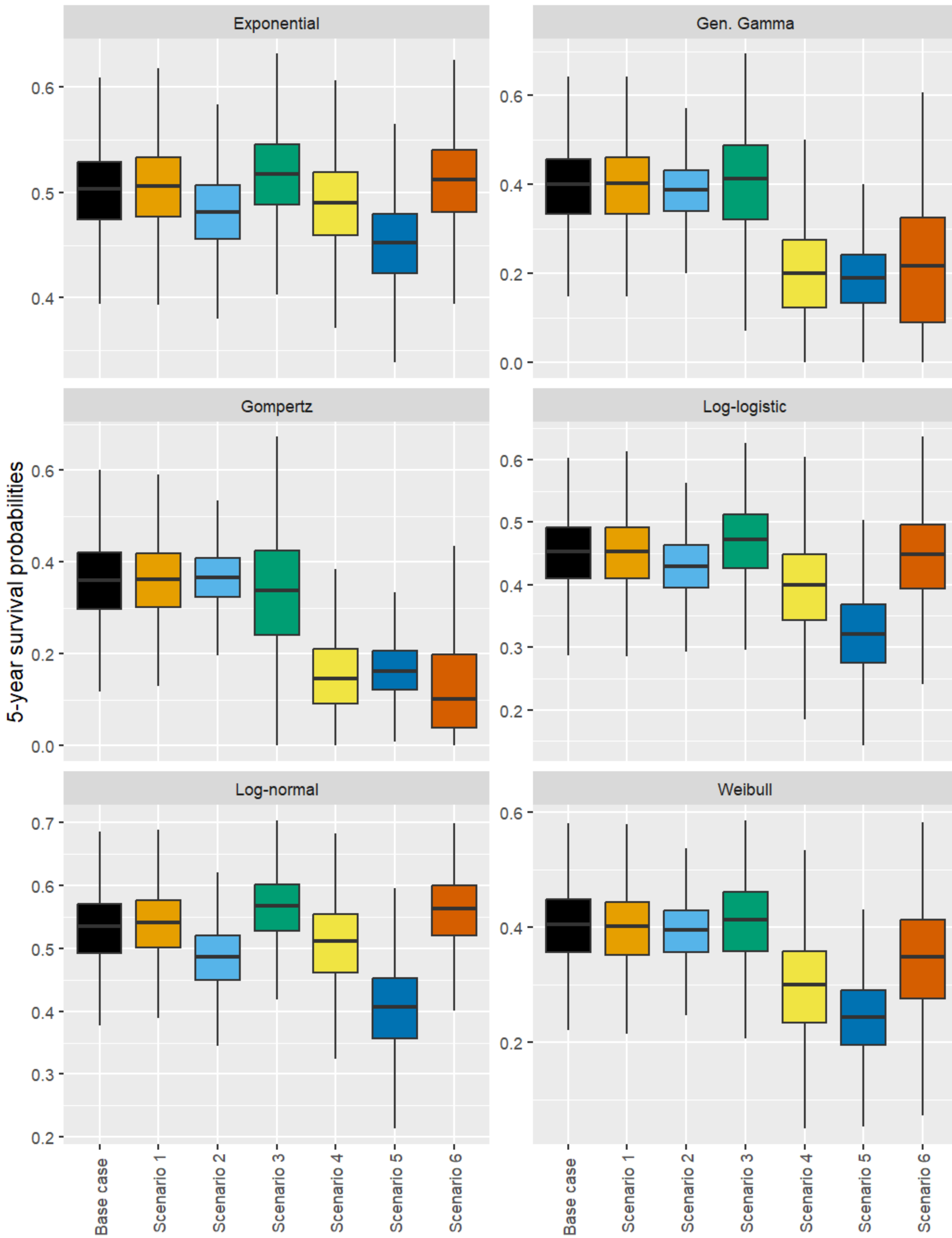

Figure 34: Boxplots of probabilistic simulations of 5-year OS in the Base Case and in Scenarios 1 through 5.

Table 19: Summary of probabilistic estimates of 5-year OS for the base case and Scenarios 1 through 5, including comparisons between each scenario and the base case (probabilistic mean difference and variance ratio)

					Scenario versus Base Case	
Distribution	Scenario	Mean	Median	CrI.95	Mean.Difference	Variance.Ratio
Exponential	Base case	0.50	0.50	(0.42, 0.58)	NA	NA
Exponential	Scenario 1	0.50	0.51	(0.42, 0.58)	0.00	1.04
Exponential	Scenario 2	0.48	0.48	(0.41, 0.55)	-0.02	0.85
Exponential	Scenario 3	0.52	0.52	(0.43, 0.59)	0.01	1.06
Exponential	Scenario 4	0.49	0.49	(0.40, 0.57)	-0.01	1.19
Exponential	Scenario 5	0.45	0.45	(0.37, 0.53)	-0.05	1.09
Exponential	Scenario 6	0.51	0.51	(0.42, 0.60)	0.01	1.19
Gen. Gamma	Base case	0.39	0.40	(0.17, 0.55)	NA	NA
Gen. Gamma	Scenario 1	0.39	0.40	(0.16, 0.55)	0.00	1.03
Gen. Gamma	Scenario 2	0.38	0.39	(0.22, 0.51)	-0.01	0.56
Gen. Gamma	Scenario 3	0.40	0.41	(0.08, 0.60)	0.01	1.74
Gen. Gamma	Scenario 4	0.20	0.20	(0.00, 0.39)	-0.19	1.15
Gen. Gamma	Scenario 5	0.19	0.19	(0.04, 0.34)	-0.20	0.65
Gen. Gamma	Scenario 6	0.21	0.22	(0.00, 0.47)	-0.18	2.15
Gompertz	Base case	0.36	0.36	(0.18, 0.52)	NA	NA
Gompertz	Scenario 1	0.36	0.36	(0.19, 0.52)	0.00	0.93
Gompertz	Scenario 2	0.37	0.37	(0.24, 0.48)	0.01	0.49
Gompertz	Scenario 3	0.33	0.34	(0.08, 0.56)	-0.02	2.01
Gompertz	Scenario 4	0.16	0.15	(0.02, 0.34)	-0.20	0.88
Gompertz	Scenario 5	0.17	0.16	(0.06, 0.29)	-0.19	0.48
Gompertz	Scenario 6	0.13	0.10	(0.00, 0.39)	-0.23	1.56
Log-logistic	Base case	0.45	0.45	(0.32, 0.55)	NA	NA
Log-logistic	Scenario 1	0.45	0.45	(0.32, 0.55)	0.00	1.02
Log-logistic	Scenario 2	0.43	0.43	(0.32, 0.52)	-0.02	0.73
Log-logistic	Scenario 3	0.47	0.47	(0.32, 0.58)	0.02	1.16
Log-logistic	Scenario 4	0.39	0.40	(0.24, 0.53)	-0.05	1.63
Log-logistic	Scenario 5	0.32	0.32	(0.19, 0.45)	-0.13	1.27
Log-logistic	Scenario 6	0.44	0.45	(0.28, 0.57)	-0.01	1.60
Log-normal	Base case	0.53	0.53	(0.41, 0.63)	NA	NA
Log-normal	Scenario 1	0.54	0.54	(0.41, 0.63)	0.01	1.00
Log-normal	Scenario 2	0.48	0.49	(0.37, 0.58)	-0.05	0.84
Log-normal	Scenario 3	0.56	0.57	(0.44, 0.66)	0.03	0.95
Log-normal	Scenario 4	0.50	0.51	(0.35, 0.62)	-0.02	1.45
Log-normal	Scenario 5	0.40	0.41	(0.26, 0.53)	-0.13	1.53
Log-normal	Scenario 6	0.56	0.56	(0.42, 0.66)	0.03	1.10
Weibull	Base case	0.40	0.41	(0.26, 0.51)	NA	NA
Weibull	Scenario 1	0.40	0.40	(0.25, 0.52)	0.00	1.09
Weibull	Scenario 2	0.39	0.40	(0.27, 0.49)	-0.01	0.68
Weibull	Scenario 3	0.41	0.41	(0.24, 0.53)	0.01	1.29
Weibull	Scenario 4	0.29	0.30	(0.11, 0.46)	-0.11	1.79
Weibull	Scenario 5	0.24	0.24	(0.11, 0.37)	-0.16	1.06
Weibull	Scenario 6	0.34	0.35	(0.13, 0.51)	-0.06	2.21

6 Assessment of multivariate normal approximation of joint parameter distributions

Table 20: Comparison of probabilistic estimates Mean OS (months) and 5-year OS probabilities using the multivariate normal approximation and the weighted samples obtained from the importance sampling algorithm

		MVN approximation			Posterior Samples		
Distribution	Output	Mean	Median	95% CrI	Mean	Median	95% CrI
Exponential	5-year OS (prob.)	0.50	0.50	(0.42 to 0.58)	0.50	0.50	(0.42 to 0.58)
Exponential	Mean OS (months)	87.69	87.07	(69.33 to 109.25)	87.84	87.07	(70.01 to 109.64)
Gen. Gamma	5-year OS (prob.)	0.39	0.40	(0.18 to 0.55)	0.39	0.39	(0.22 to 0.56)
Gen. Gamma	Mean OS (months)	70.82	56.92	(39.19 to 208.51)	70.26	56.35	(41.27 to 211.28)
Gompertz	5-year OS (prob.)	0.35	0.36	(0.18 to 0.51)	0.35	0.35	(0.18 to 0.52)
Gompertz	Mean OS (months)	53.66	49.65	(39.41 to 84.54)	55.23	49.29	(39.36 to 94.65)
Log-logistic	5-year OS (prob.)	0.45	0.45	(0.31 to 0.56)	0.45	0.45	(0.33 to 0.56)
Log-logistic	Mean OS (months)	119.55	116.67	(69.74 to 186.52)	119.94	116.88	(71.94 to 187.36)
Log-normal	5-year OS (prob.)	0.53	0.53	(0.41 to 0.63)	0.53	0.53	(0.41 to 0.63)
Log-normal	Mean OS (months)	184.78	180.70	(99.17 to 293.62)	183.54	179.98	(100.61 to 294.28)
Weibull	5-year OS (prob.)	0.40	0.41	(0.25 to 0.52)	0.40	0.40	(0.26 to 0.52)
Weibull	Mean OS (months)	61.86	60.94	(43.28 to 85.75)	61.73	60.27	(44.42 to 87.86)

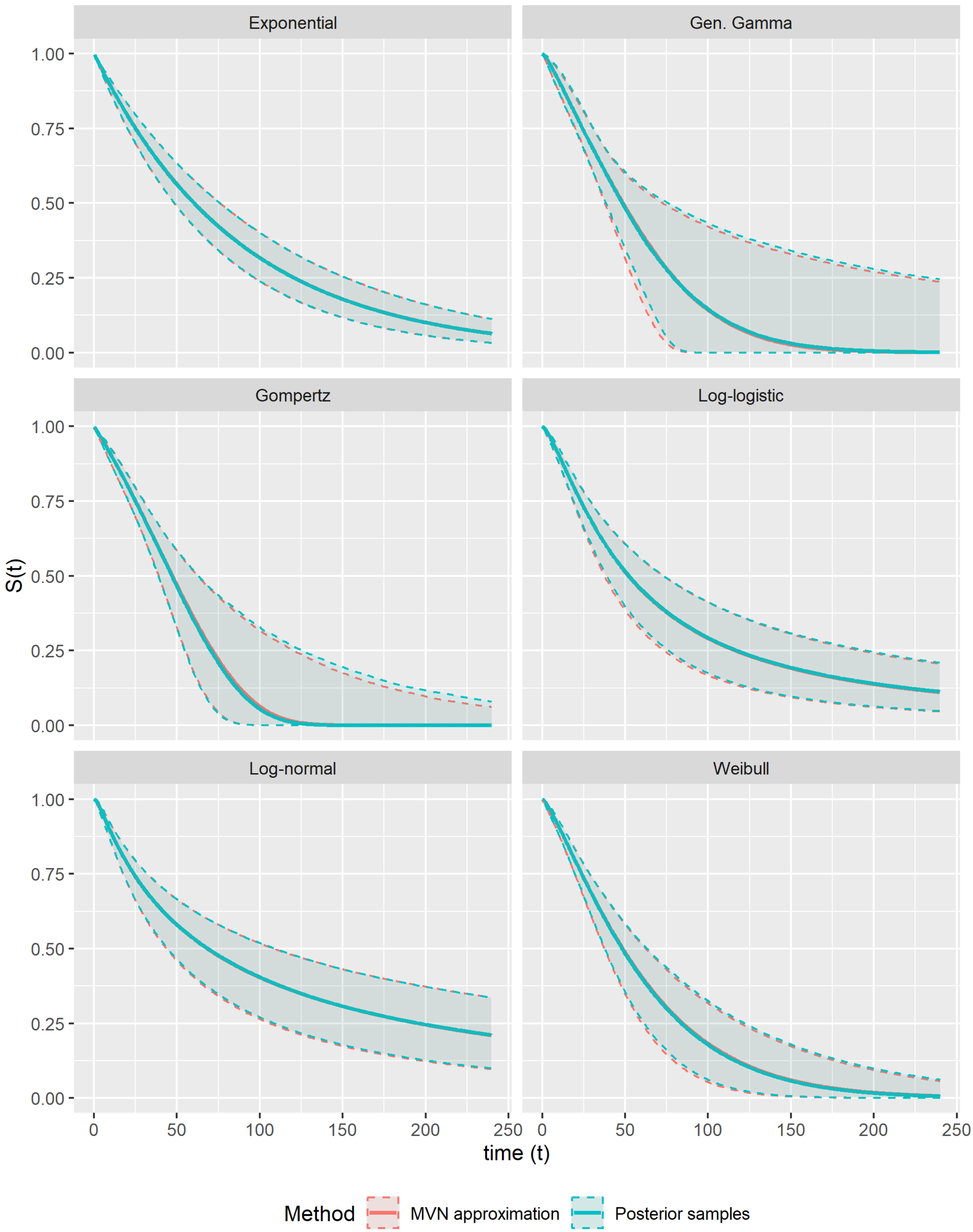

Figure 35: Survival curves (posterior median and 95 percent credible intervals) obtained from importance sampling-weighted analysis and from the multivariate normal approximation of the posterior parameter distribution, Exponential distribution

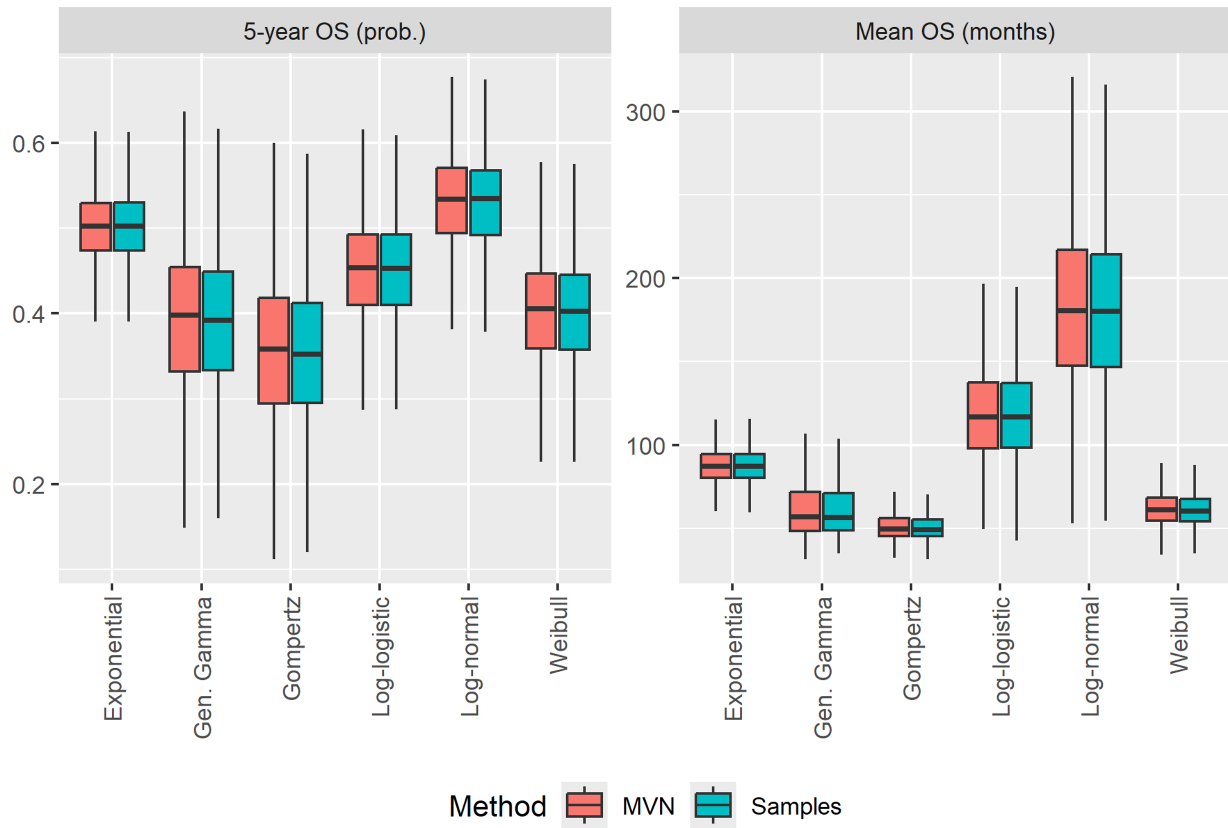

Figure 36: Boxplots of probabilistic simulations of Mean OS (months) and 5-year OS probabilities using the multivariate normal approximation and the weighted samples obtained from the importance sampling algorithm.

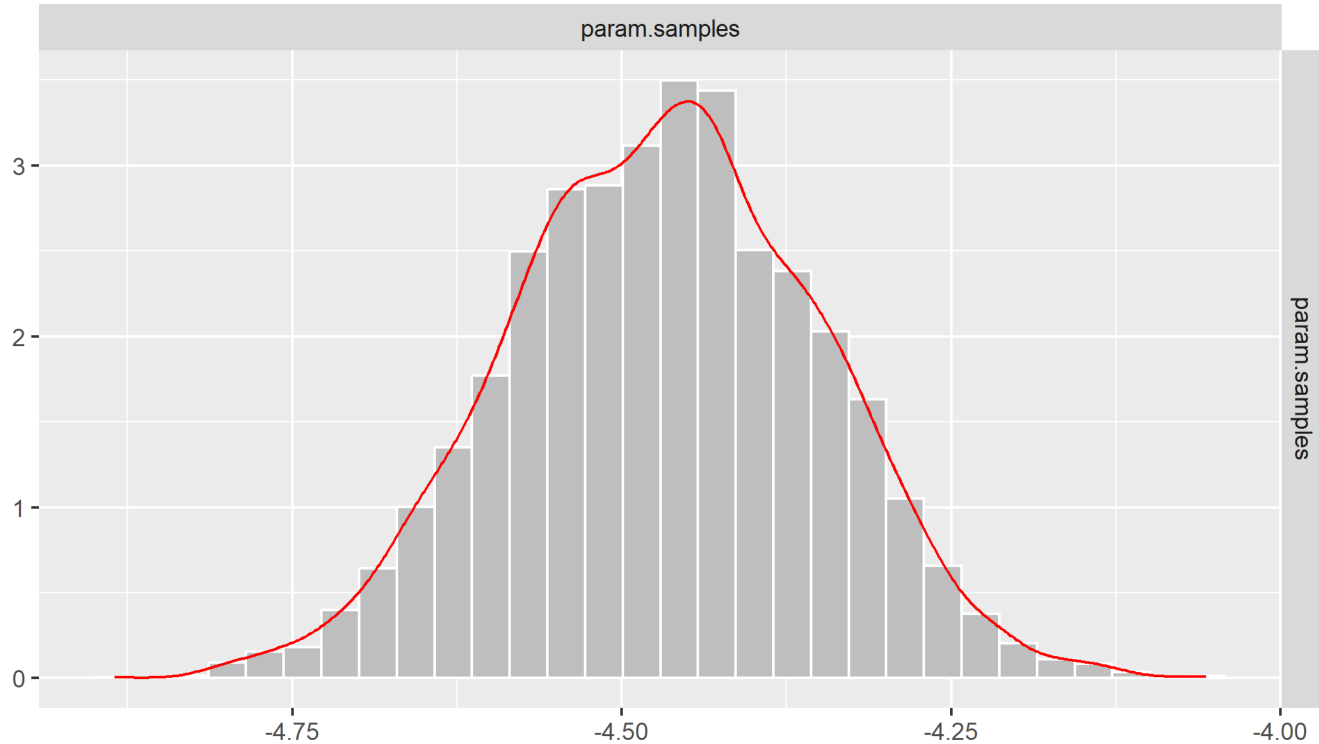

Figure 37: Plots of samples from the posterior distribution obtained from the importance sampling algorithm, Exponential distribution

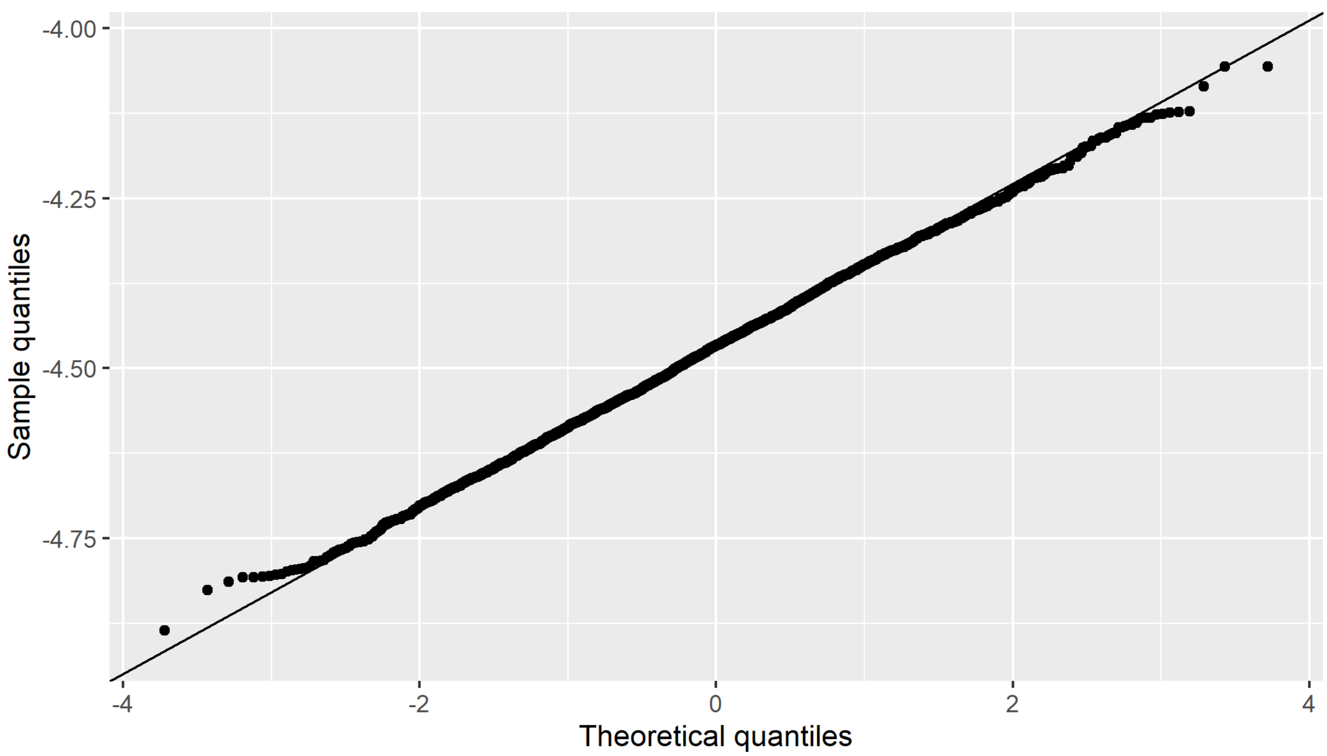

Figure 38: QQ plot to assess normality of the posterior samples obtained from the importance sampling algorithm, Exponential distribution

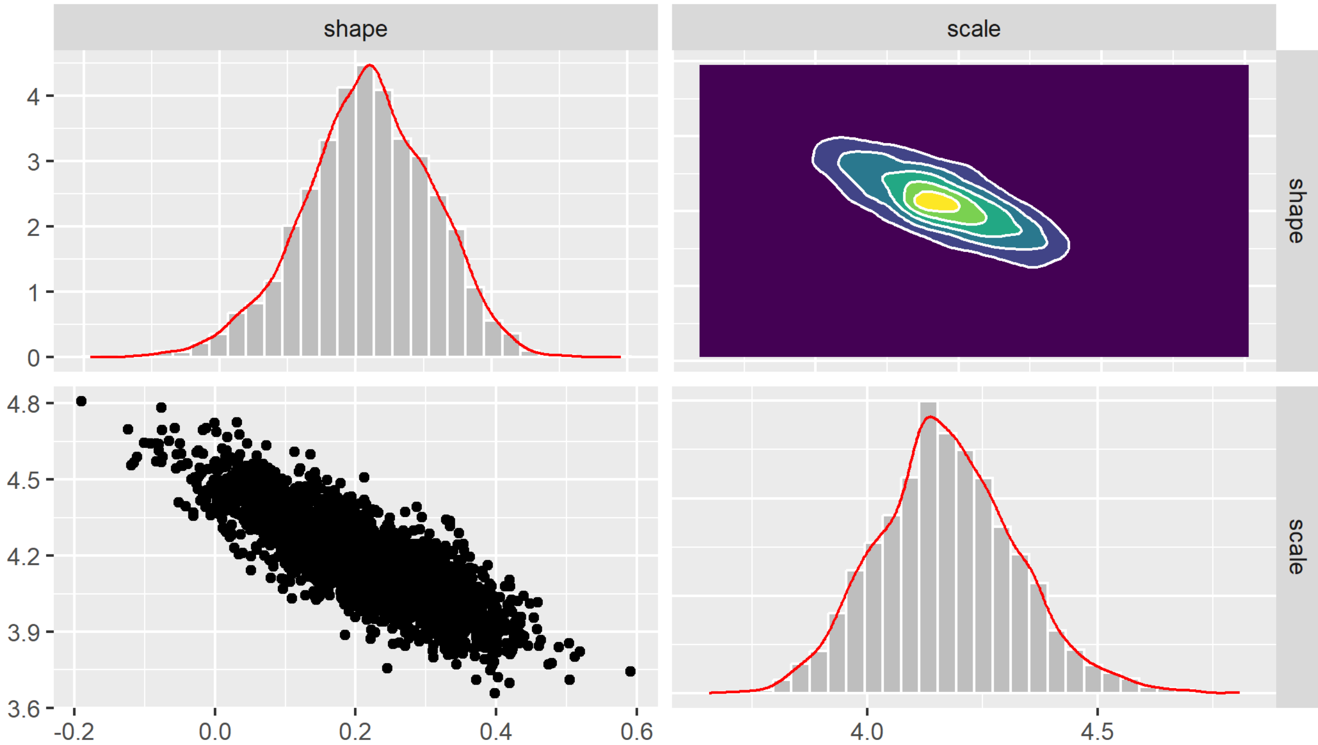

Figure 39: Plots of samples from the posterior distribution obtained from the importance sampling algorithm, Weibull distribution

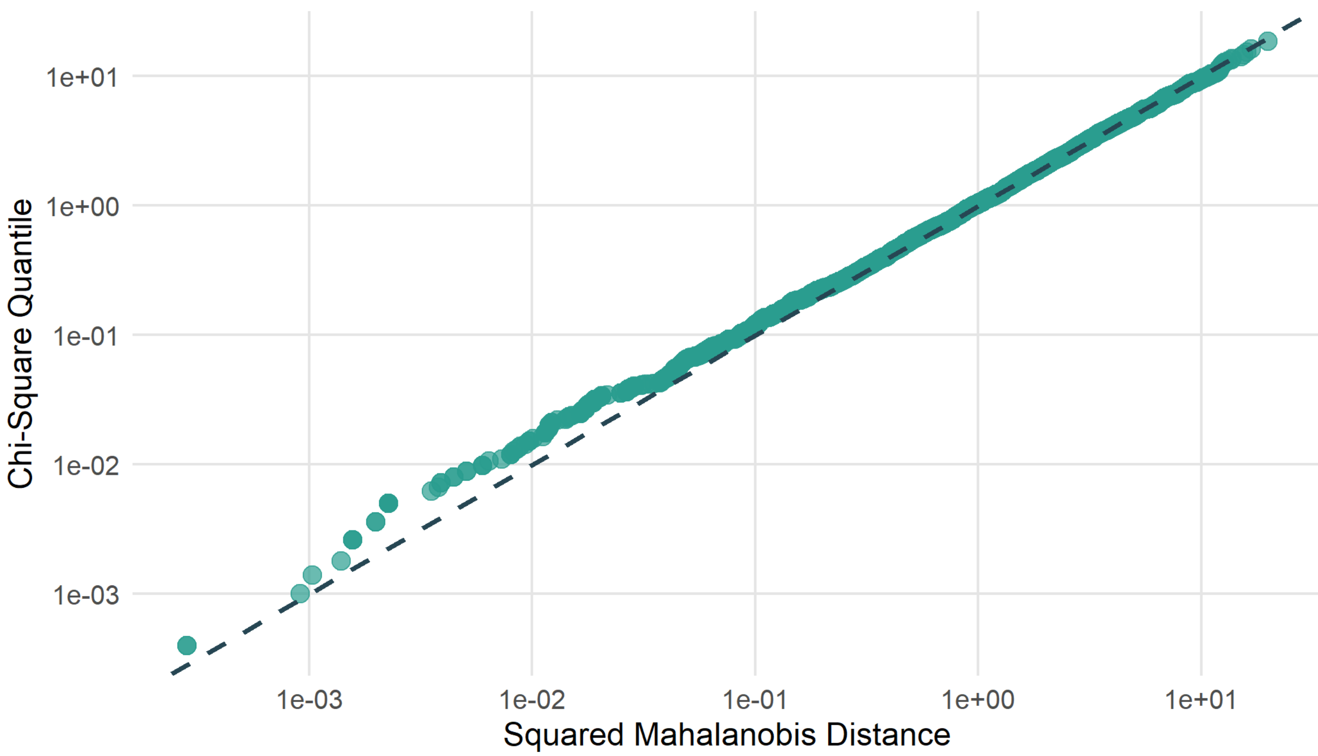

Figure 40: QQ plot to assess normality of the posterior samples obtained from the importance sampling algorithm, Weibull distribution

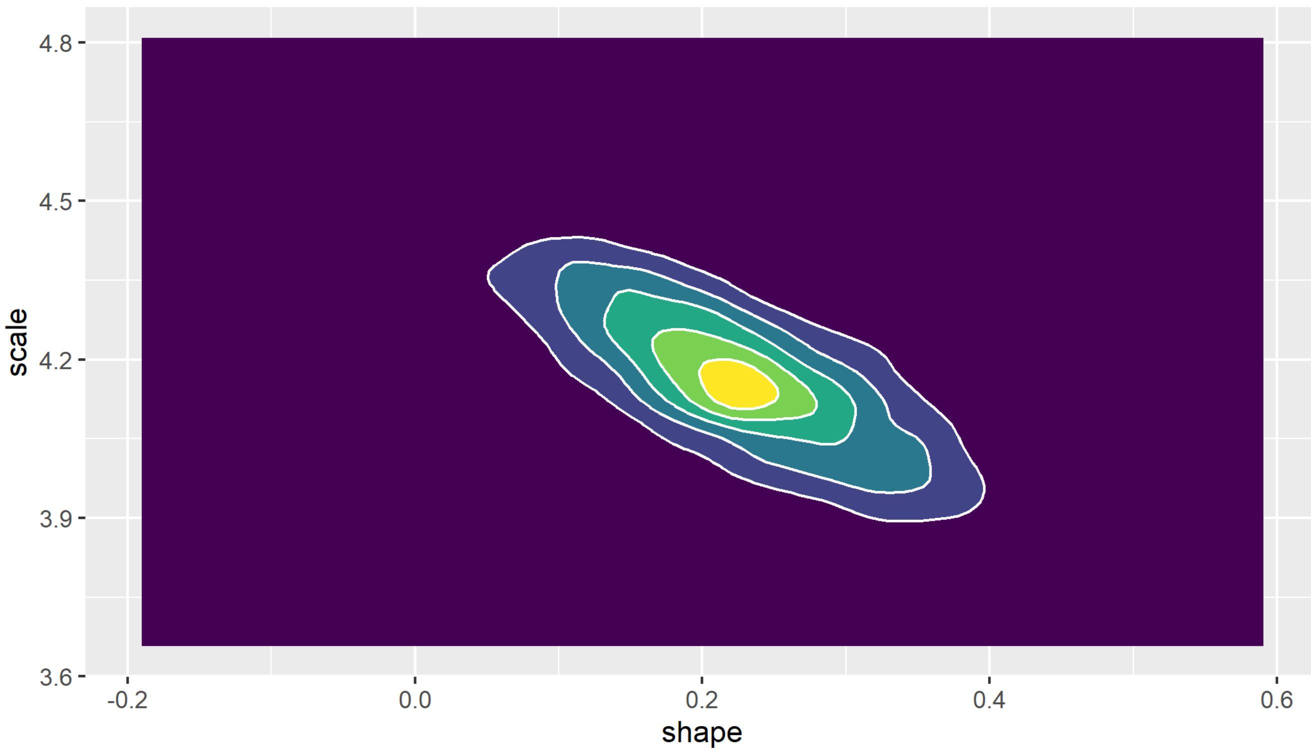

Figure 41: Contour plot of the posterior parameter samples obtained from the importance sampling algorithm, Weibull distribution

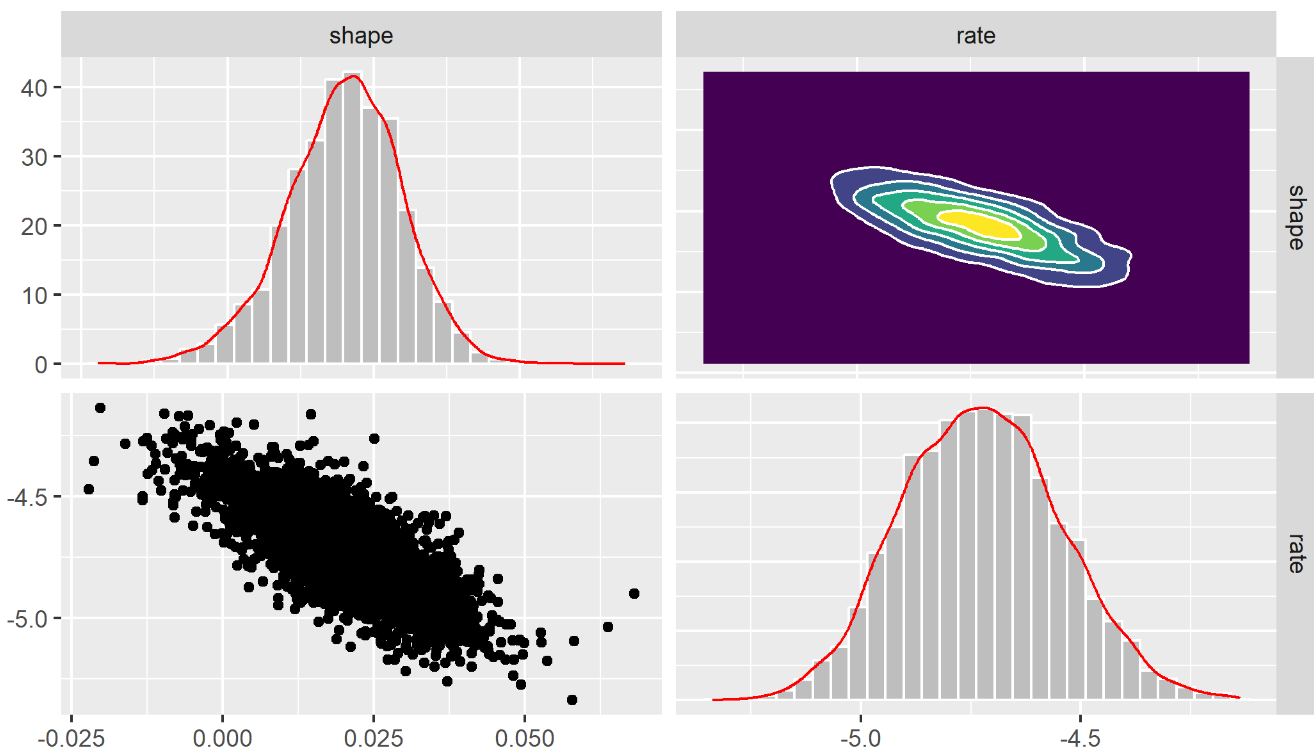

Figure 42: Plots of samples from the posterior distribution obtained from the importance sampling algorithm, Gompertz distribution

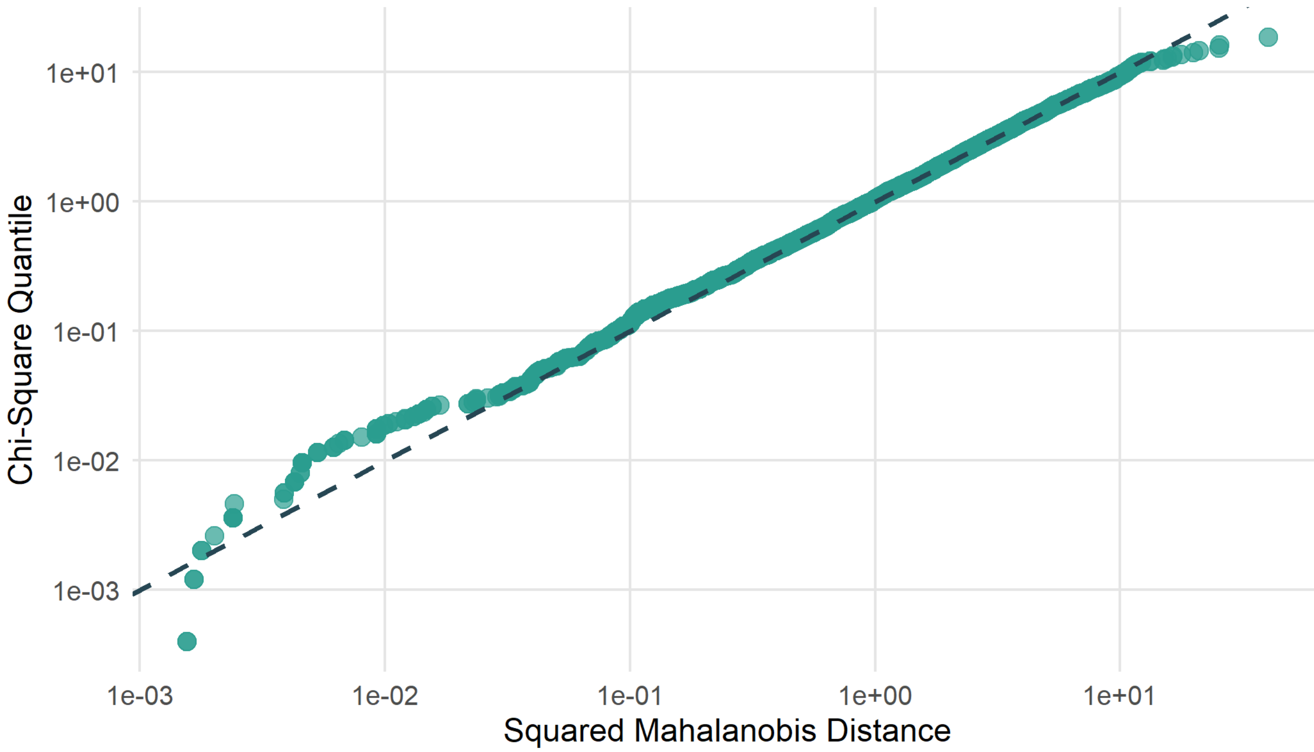

Figure 43: QQ plot to assess normality of the posterior samples obtained from the importance sampling algorithm, Gompertz distribution

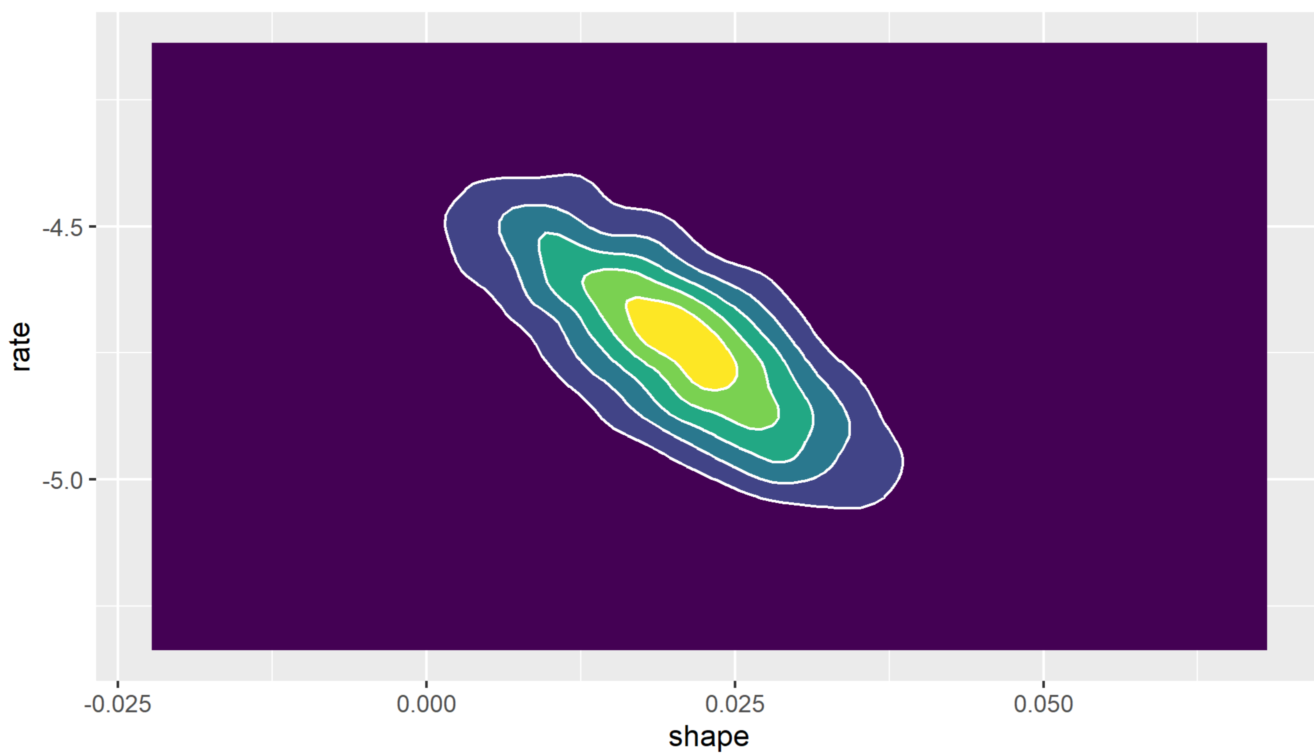

Figure 44: Contour plot of the posterior parameter samples obtained from the importance sampling algorithm, Gompertz distribution

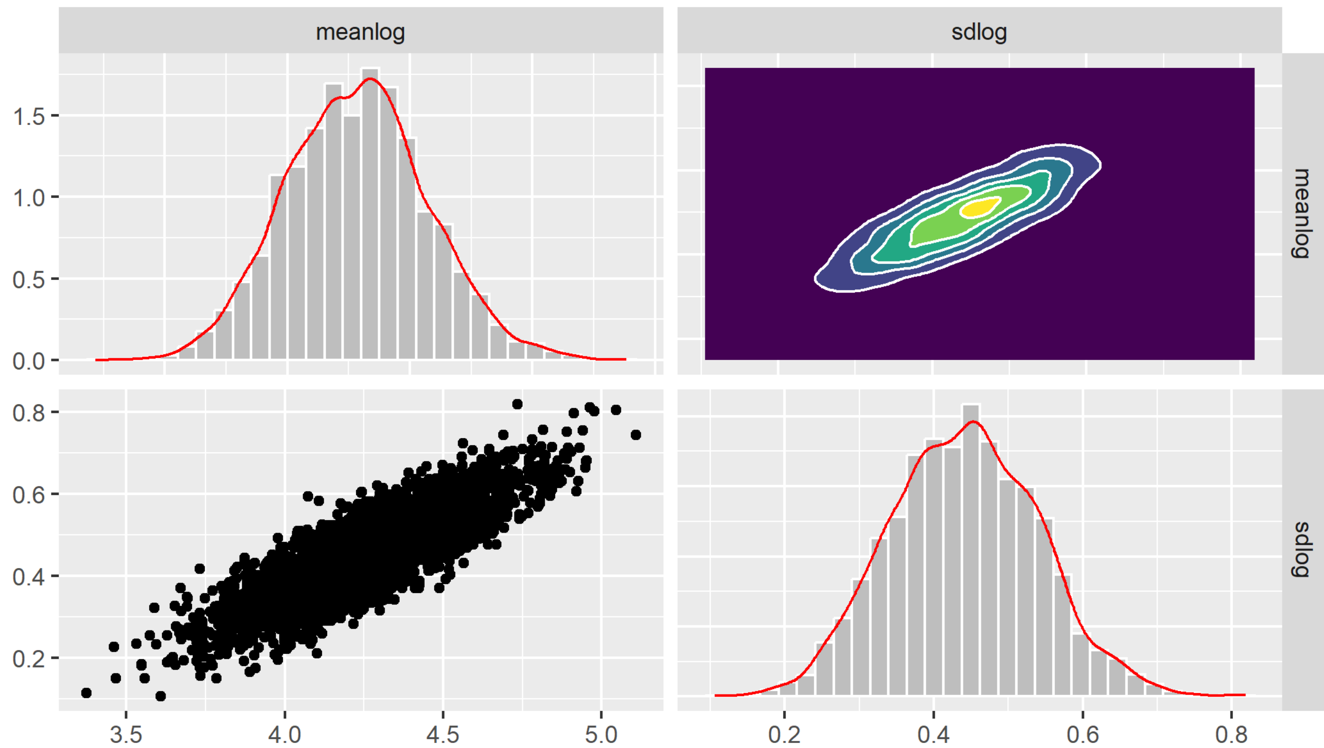

Figure 45: Plots of samples from the posterior distribution obtained from the importance sampling algorithm, Log-normal distribution

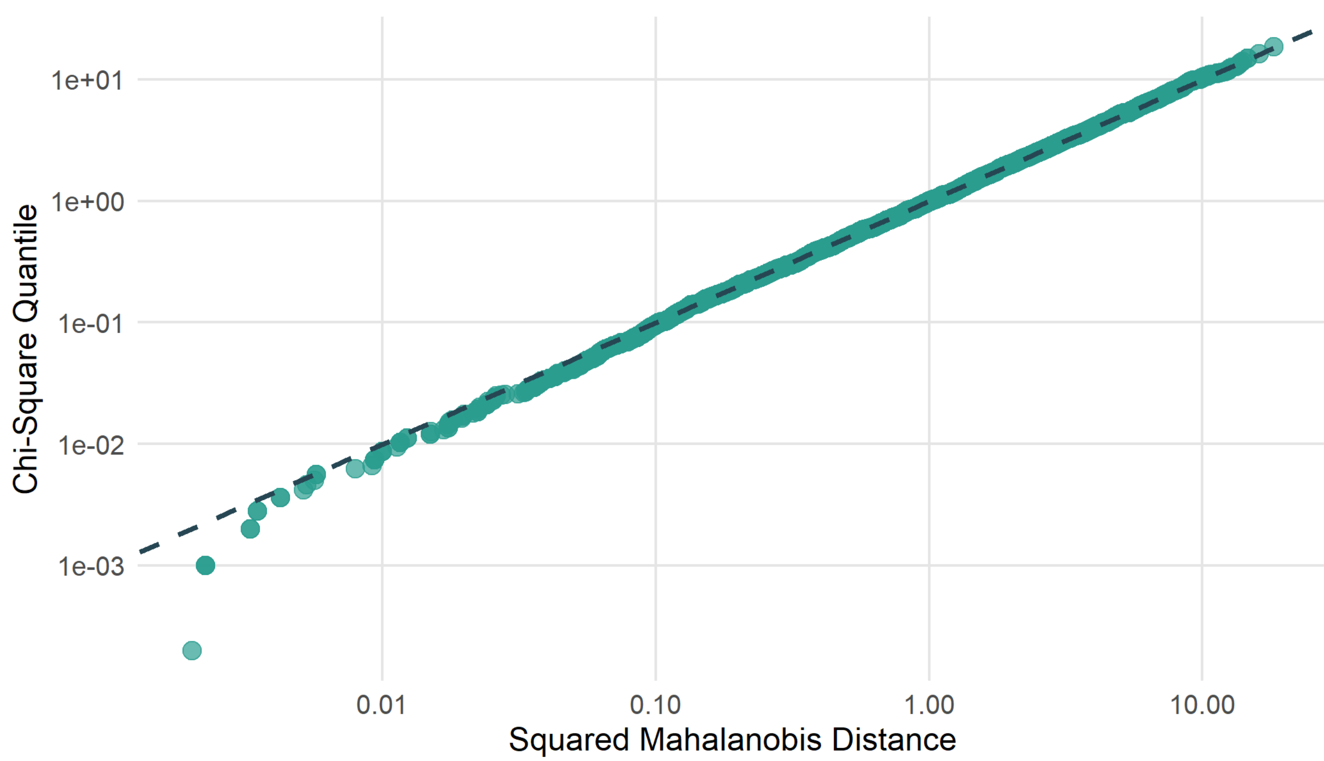

Figure 46: QQ plot to assess normality of the posterior samples obtained from the importance sampling algorithm, Log-normal distribution

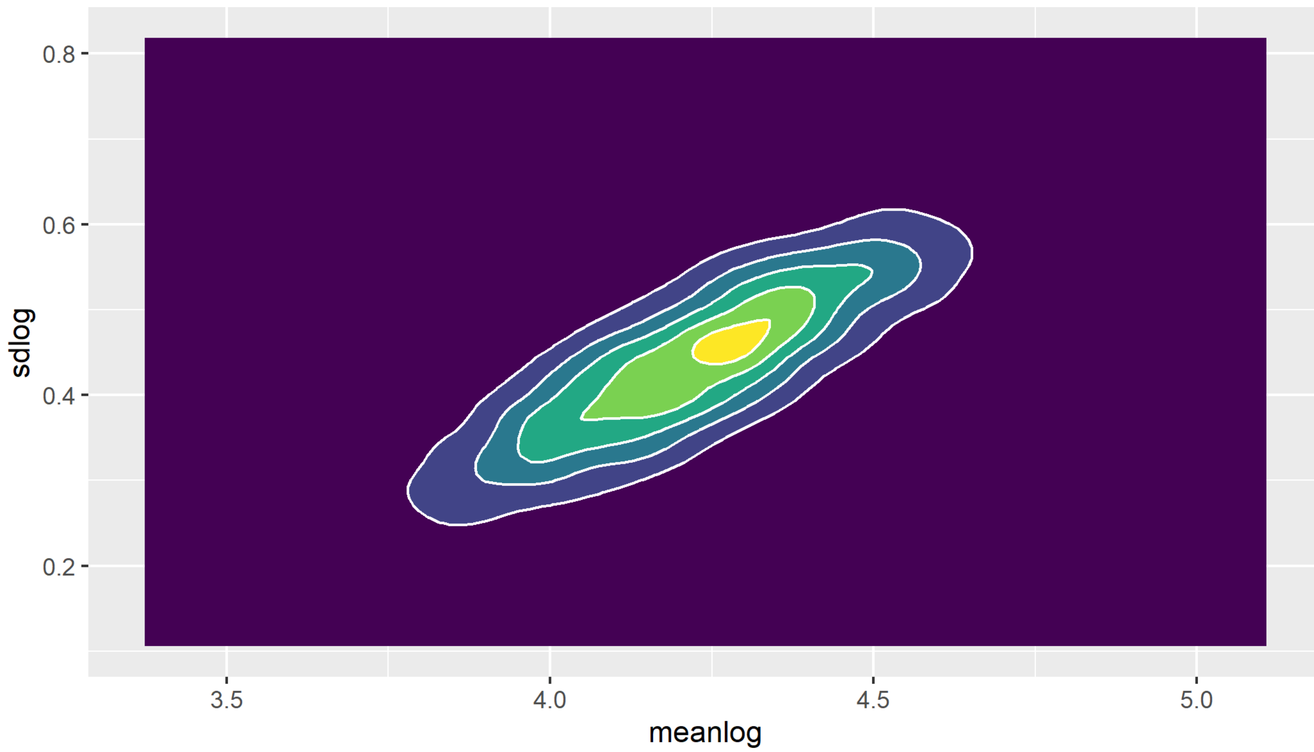

Figure 47: Contour plot of the posterior parameter samples obtained from the importance sampling algorithm, Log-normal distribution

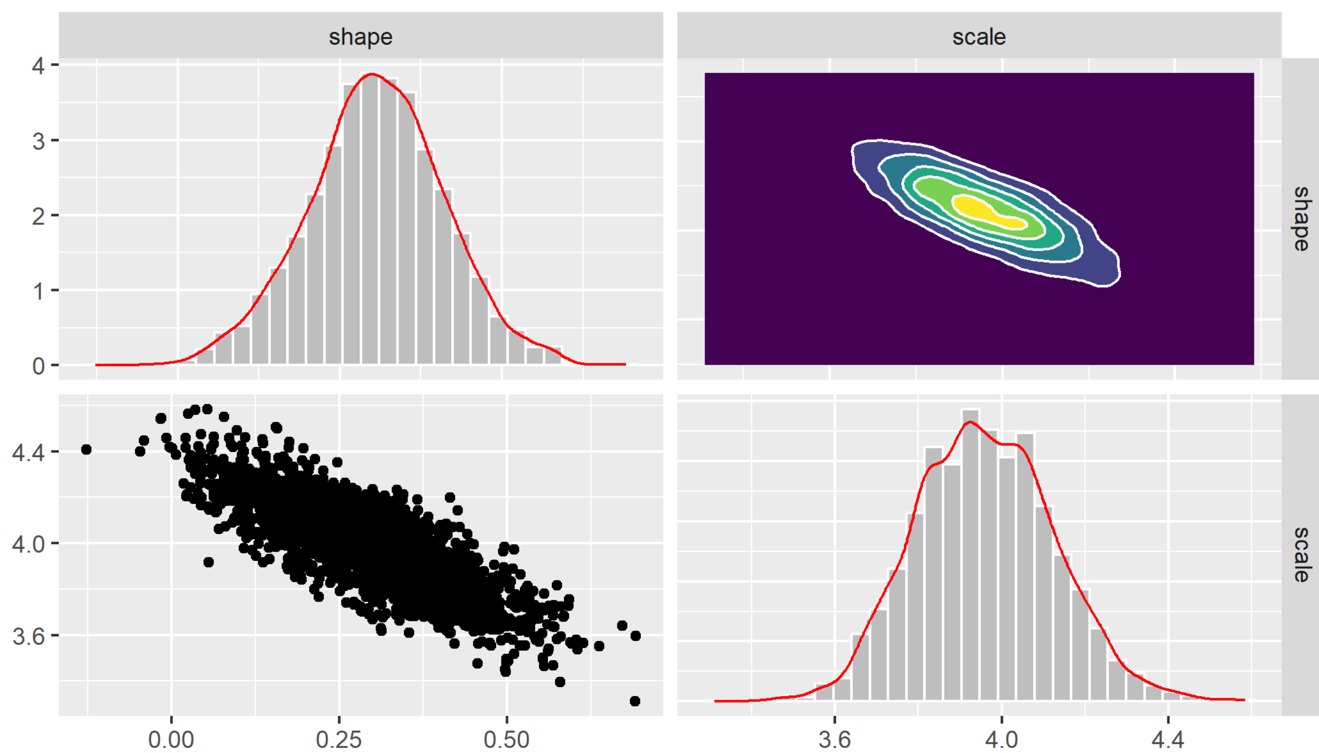

Figure 48: Plots of samples from the posterior distribution obtained from the importance sampling algorithm, Log-logistic distribution

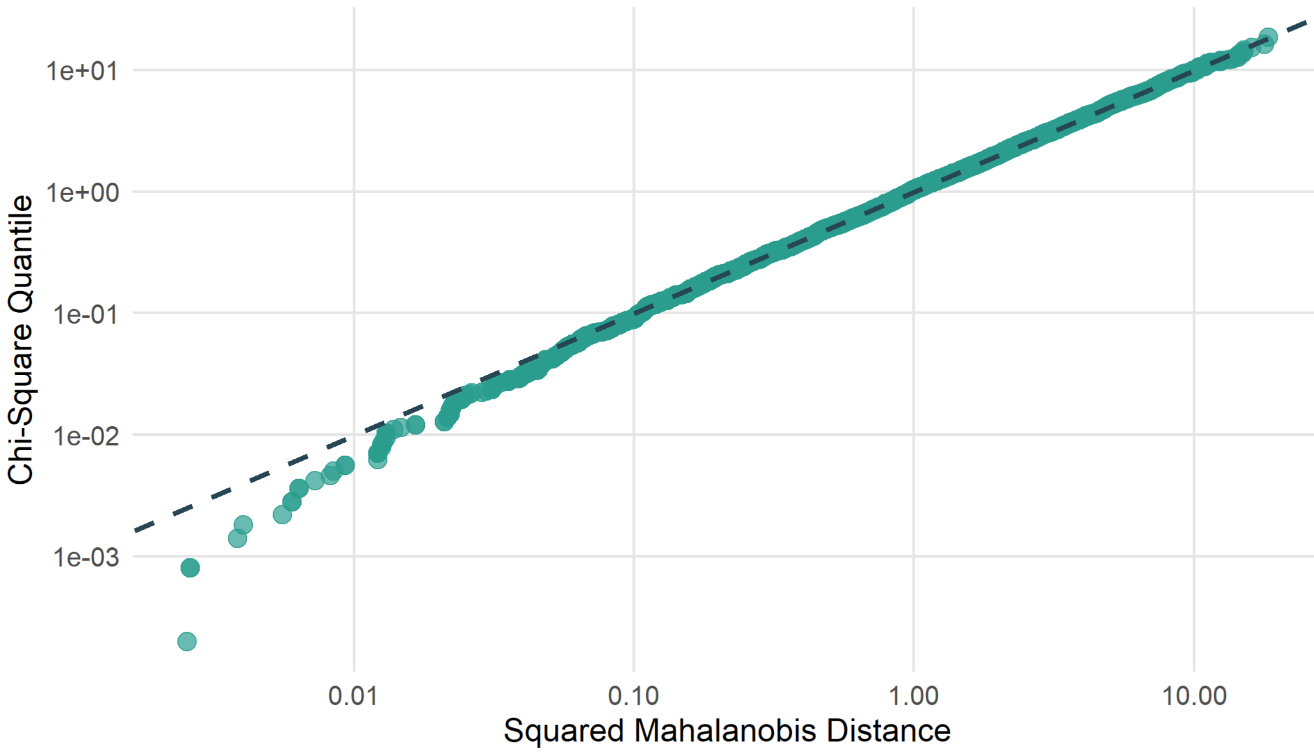

Figure 49: QQ plot to assess normality of the posterior samples obtained from the importance sampling algorithm, Log-logistic distribution

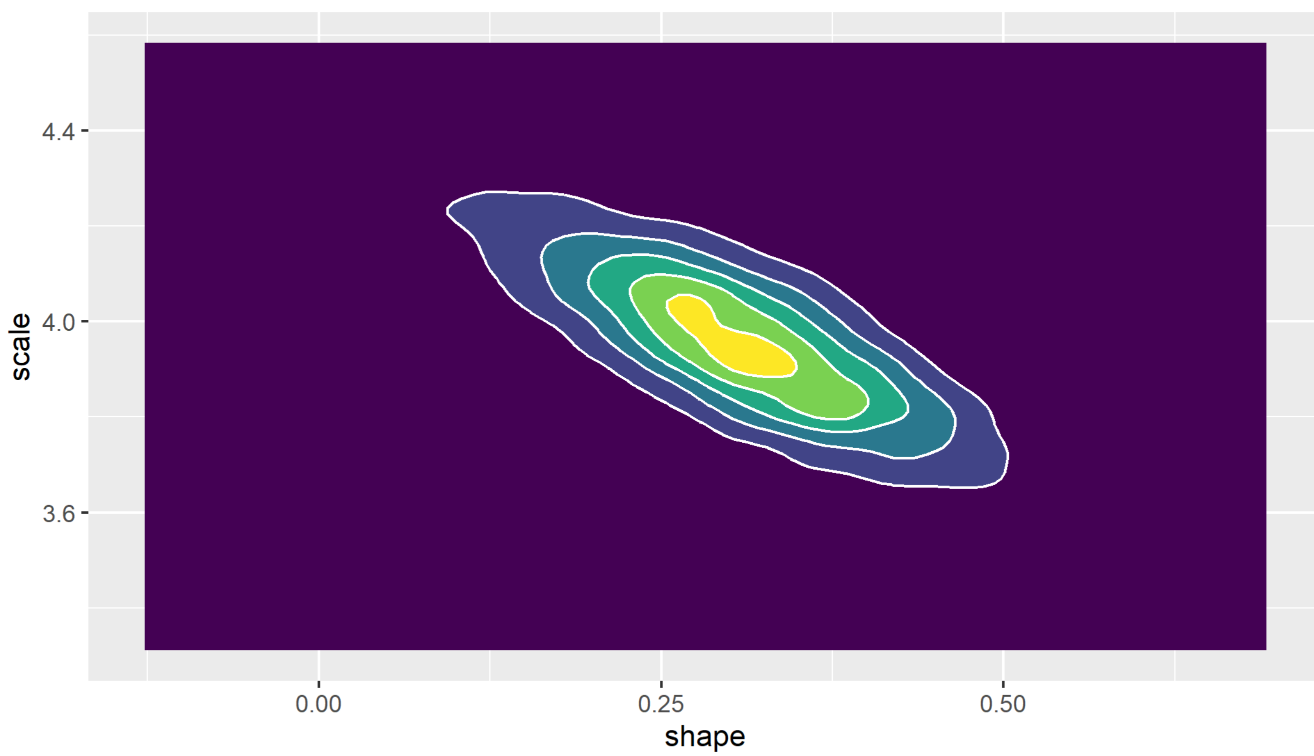

Figure 50: Contour plot of the posterior parameter samples obtained from the importance sampling algorithm, Log-logistic distribution

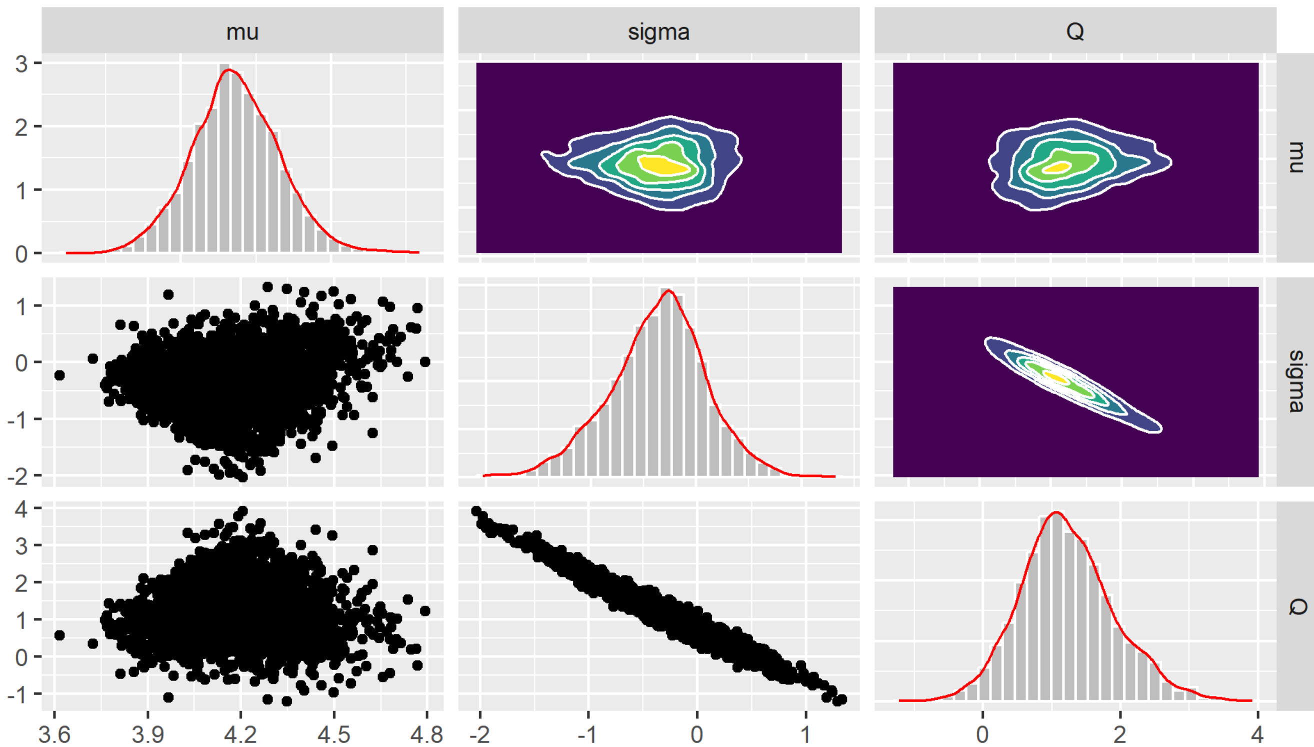

Figure 51: Plots of samples from the posterior distribution obtained from the importance sampling algorithm, Gen. Gamma distribution

Figure 52: QQ plot to assess normality of the posterior samples obtained from the importance sampling algorithm, Gen. Gamma distribution

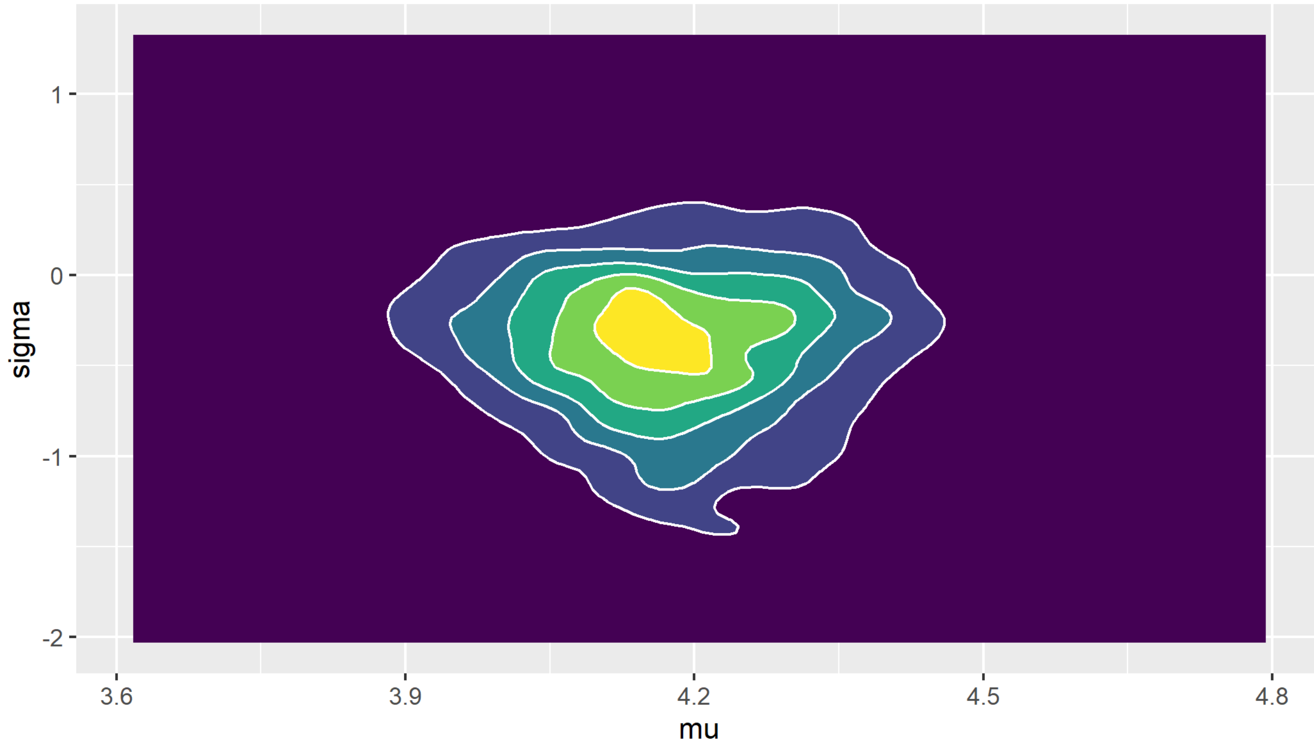

Figure 53: Contour plot of the posterior parameter samples obtained from the importance sampling algorithm, Gen. Gamma distribution, parameters mu and sigma (marginalised over Q)

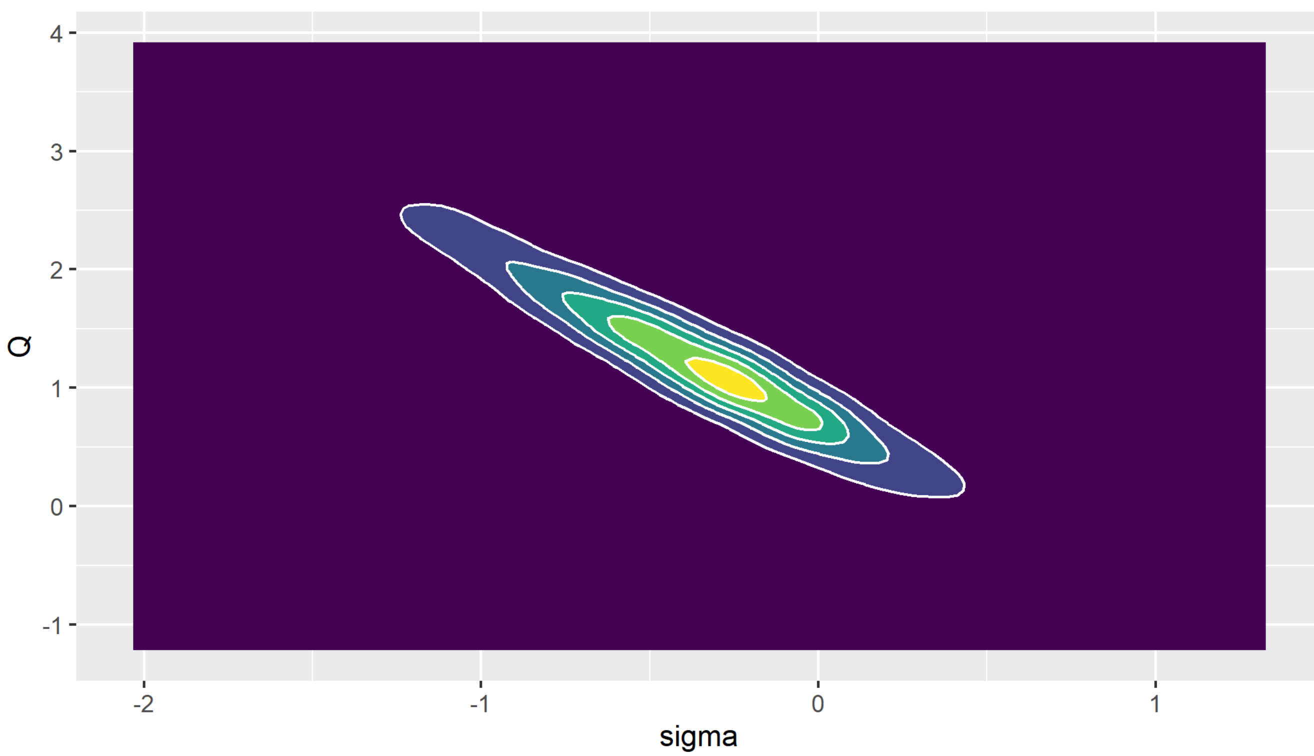

Figure 54: Contour plot of the posterior parameter samples obtained from the importance sampling algorithm, Gen. Gamma distribution, parameters sigma and Q (marginalised over mu)

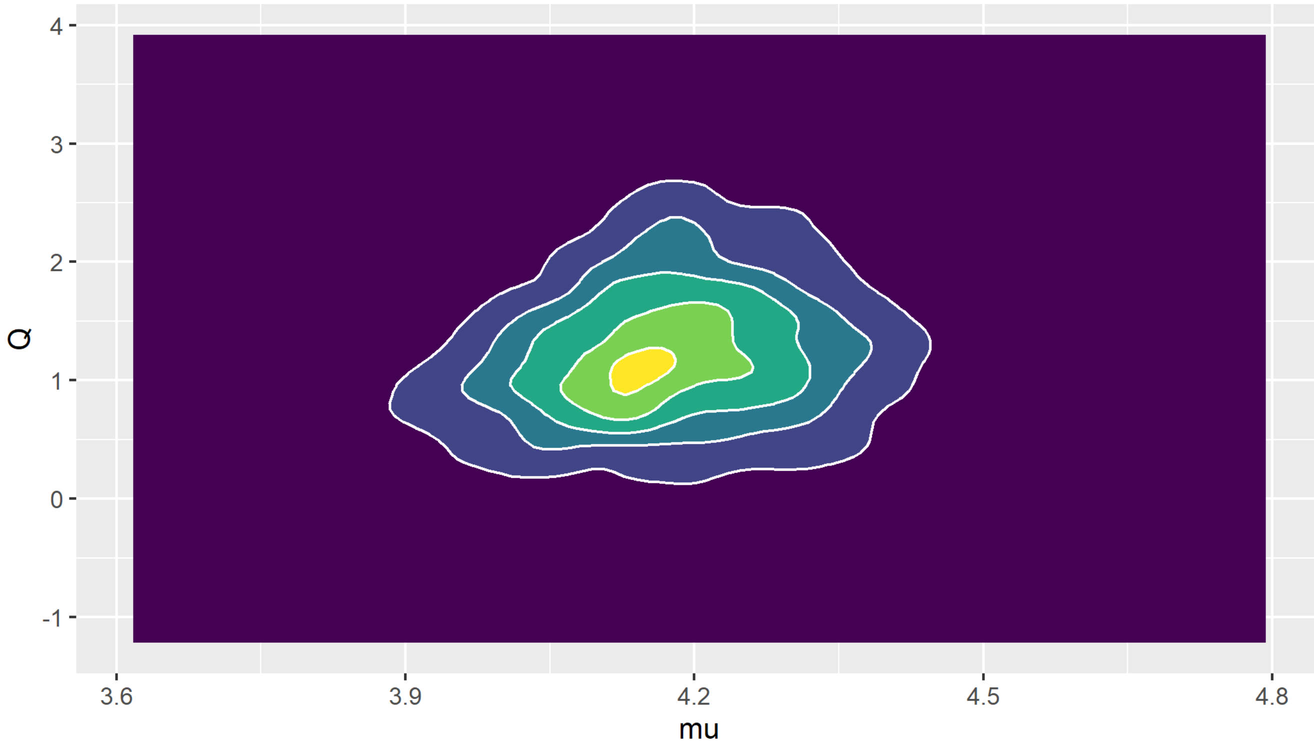

Figure 55: Contour plot of the posterior parameter samples obtained from the importance sampling algorithm, Gen. Gamma distribution, parameters mu and Q (marginalised over sigma)